

positive; also it is *not* finitely integrable at ∞ . Therefore, as x travels from 0 to ∞ , we see that $f(x)$ increases from 0 to ∞ , thus f maps $[0, \infty)$ to $[0, \infty)$. On the other hand, when $z = x$ is negative, then

$$f'(z) = \alpha|x|^{\alpha-1}e^{i\pi(\alpha-1)} = -\alpha|x|^{\alpha-1}e^{i\pi\alpha},$$

so f maps the segment $(-\infty, 0]$ to $(e^{i\pi\alpha}\infty, 0]$. The situation is illustrated in Figure 3 where the infinite segment A is mapped to A' and the segment B is mapped to B' , with the direction of travel indicated in Figure 3.



Figure 3. The conformal map z^α

EXAMPLE 2. Next, we consider for $z \in \mathbb{H}$,

$$f(z) = \int_0^z \frac{d\zeta}{(1-\zeta^2)^{1/2}},$$

where the integral is taken from 0 to z along any path in the closed upper half-plane. We choose the branch for $(1-\zeta^2)^{1/2}$ that makes it holomorphic in the upper half-plane and positive when $-1 < \zeta < 1$. As a result

$$(1-\zeta^2)^{-1/2} = i(\zeta^2-1)^{-1/2} \quad \text{when } \zeta > 1.$$

We observe that f maps the real line to the boundary of the half-strip pictured in Figure 4.

In fact, since $f(\pm 1) = \pm\pi/2$, and $f'(x) > 0$ if $-1 < x < 1$, we see that f maps the segment B to B' . Moreover

$$f(x) = \frac{\pi}{2} + \int_1^x f'(x) dx \quad \text{when } x > 1, \quad \text{and} \quad \int_1^\infty \frac{dx}{(x^2-1)^{1/2}} = \infty.$$

Thus, as x travels along the segment C , the image traverses the infinite segment C' . Similarly segment A is mapped to A' .

Note the connection of this example with Example 8 in Section 1.2. In fact, one can show that the function $f(z)$ is the inverse to the function

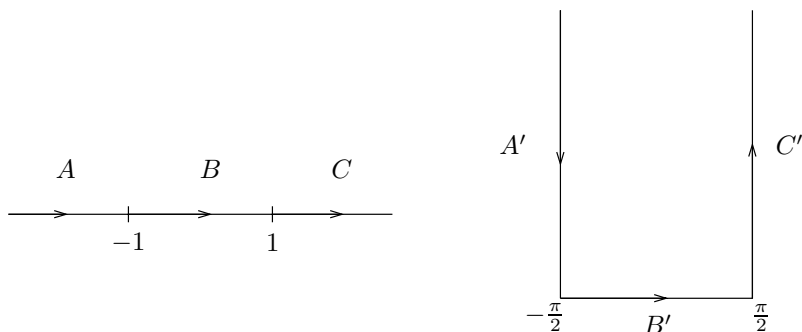


Figure 4. Mapping of the boundary in Example 2

$\sin z$, and hence f takes $\mathbb{H}$ conformally to the interior of the half-strip bounded by the segments A' , B' , and C' .

EXAMPLE 3. Here we take

$$f(z) = \int_0^z \frac{d\zeta}{[(1 - \zeta^2)(1 - k^2\zeta^2)]^{1/2}}, \quad z \in \mathbb{H},$$

where k is a fixed real number with $0 < k < 1$ (the branch of $[(1 - \zeta^2)(1 - k^2\zeta^2)]^{1/2}$ in the upper half-plane is chosen to be the one that is positive when ζ is real and $-1 < \zeta < 1$). Integrals of this kind are called **elliptic integrals**, because variants of these arise in the calculation of the arc-length of an ellipse. We shall observe that f maps the real axis onto the rectangle shown in Figure 5(b), where K and K' are determined by

$$K = \int_0^1 \frac{dx}{[(1 - x^2)(1 - k^2x^2)]^{1/2}}, \quad K' = \int_1^{1/k} \frac{dx}{[(x^2 - 1)(1 - k^2x^2)]^{1/2}}.$$

We divide the real axis into four “segments,” with division points $-1/k$, -1 , 1 , and $1/k$ (see Figure 5(a)). The segments are $[-1/k, -1]$, $[-1, 1]$, $[1, 1/k]$, and $[1/k, -1/k]$, the last consisting of the union of the two half-segments $[1/k, \infty)$ and $(-\infty, -1/k]$. It is clear from the definitions that $f(\pm 1) = \pm K$, and since $f'(x) > 0$, when $-1 < x < 1$, it follows that f maps the segment $[-1, 1]$ to $[-K, K]$. Moreover, since

$$f(z) = K + \int_1^x \frac{d\zeta}{[(1 - \zeta^2)(1 - k^2\zeta^2)]^{1/2}} \quad \text{if } 1 < x < 1/k,$$

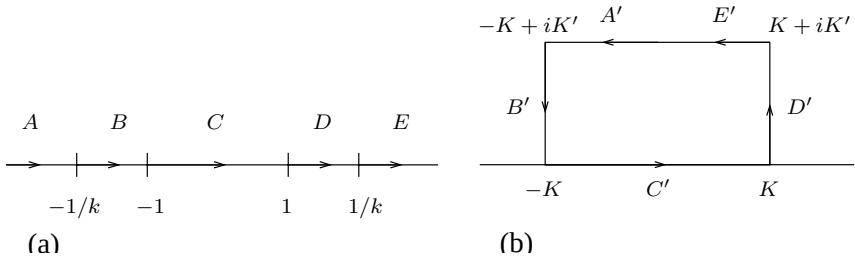


Figure 5. Mapping of the boundary in Example 3

we see that f maps the segment $[1, 1/k]$ to $[K, K + iK']$, where K' was defined above. Similarly, f maps $[-1/k, -1]$ to $[-K + iK', -K]$. Next, when $x > 1/k$ we have

$$f'(x) = -\frac{1}{[(x^2 - 1)(k^2x^2 - 1)]^{1/2}},$$

and therefore,

$$f(x) = K + iK' - \int_{1/k}^x \frac{dx}{[(x^2 - 1)(k^2x^2 - 1)]^{1/2}}.$$

However,

$$\int_{1/k}^{\infty} \frac{dx}{[(x^2 - 1)(k^2x^2 - 1)]^{1/2}} = \int_0^1 \frac{dx}{[(1 - x^2)(1 - k^2x^2)]^{1/2}},$$

as can be seen by making the change of variables $x = 1/ku$ in the integral on the left. Thus f maps the segment $[1/k, \infty)$ to the segment $[K + iK', iK')$. Similarly f maps $(-\infty, -1/k]$ to $[-K + iK', iK')$. Altogether, then, f maps the real axis to the above rectangle, with the point at infinity corresponding to the mid-point of the upper side of the rectangle.

The results obtained so far lead naturally to two problems.

The first, which we pursue next, consists of a generalization of the above examples. More precisely we define the Schwarz-Christoffel integral and prove that it maps the real line to a polygonal line.

Second, we note that in the examples above little was inferred about the behavior of f in $\mathbb{H}$ itself. In particular, we have not shown that f maps $\mathbb{H}$ conformally to the interior of the corresponding polygon. After a careful study of the boundary behavior of conformal maps, we

prove a theorem that guarantees that the conformal map from the upper half-plane to a simply connected region bounded by a polygonal line is essentially given by a Schwarz-Christoffel integral.

4.2 The Schwarz-Christoffel integral

With the examples of the previous section in mind, we define the general **Schwarz-Christoffel integral** by

$$(5) \quad S(z) = \int_0^z \frac{d\zeta}{(\zeta - A_1)^{\beta_1} \cdots (\zeta - A_n)^{\beta_n}}.$$

Here $A_1 < A_2 < \cdots < A_n$ are n distinct points on the real axis arranged in increasing order. The exponents β_k will be assumed to satisfy the conditions $\beta_k < 1$ for each k and $1 < \sum_{k=1}^n \beta_k$.⁶

The integrand in (5) is defined as follows: $(z - A_k)^{\beta_k}$ is that branch (defined in the complex plane slit along the infinite ray $\{A_k + iy : y \leq 0\}$) which is positive when $z = x$ is real and $x > A_k$. As a result

$$(z - A_k)^{\beta_k} = \begin{cases} (x - A_k)^{\beta_k} & \text{if } x \text{ is real and } x > A_k, \\ |x - A_k|^{\beta_k} e^{i\pi\beta_k} & \text{if } x \text{ is real and } x < A_k. \end{cases}$$

The complex plane slit along the union of the rays $\cup_{k=1}^n \{A_k + iy : y \leq 0\}$ is simply connected (see Exercise 19), so the integral that defines $S(z)$ is holomorphic in this open set. Since the requirement $\beta_k < 1$ implies that the singularities $(\zeta - A_k)^{-\beta_k}$ are integrable near A_k , the function S is continuous up to the real line, including the points A_k , with $k = 1, \dots, n$. Finally, this continuity condition implies that the integral can be taken along any path in the complex plane that avoids the union of the *open* slits $\cup_{k=1}^n \{A_k + iy : y < 0\}$.

Now

$$\left| \prod_{k=1}^n (\zeta - A_k)^{-\beta_k} \right| \leq c |\zeta|^{-\sum \beta_k}$$

for $|\zeta|$ large, so the assumption $\sum \beta_k > 1$ guarantees the convergence of the integral (5) at infinity. This fact and Cauchy's theorem imply that $\lim_{r \rightarrow \infty} S(re^{i\theta})$ exists and is independent of the angle θ , $0 \leq \theta \leq \pi$. We call this limit a_∞ , and we let $a_k = S(A_k)$ for $k = 1, \dots, n$.

⁶Note that the case $\sum \beta_k \leq 1$, which occurs in Examples 1 and 2 above is excluded. However, a modification of the proposition that follows can be made to take these cases into account; but then $S(z)$ is no longer bounded in the upper half-plane.

Proposition 4.1 Suppose $S(z)$ is given by (5).

- (i) If $\sum_{k=1}^n \beta_k = 2$, and $\mathfrak{p}$ denotes the polygon whose vertices are given (in order) by $a_1, \dots, a_n$, then S maps the real axis onto $\mathfrak{p} - \{a_\infty\}$. The point a_∞ lies on the segment $[a_n, a_1]$ and is the image of the point at infinity. Moreover, the (interior) angle at the vertex a_k is $\alpha_k \pi$ where $\alpha_k = 1 - \beta_k$.
- (ii) There is a similar conclusion when $1 < \sum_{k=1}^n \beta_k < 2$, except now the image of the extended line is the polygon of $n+1$ sides with vertices $a_1, a_2, \dots, a_n, a_\infty$. The angle at the vertex a_∞ is $\alpha_\infty \pi$ with $\alpha_\infty = 1 - \beta_\infty$, where $\beta_\infty = 2 - \sum_{k=1}^n \beta_k$.

Figure 6 illustrates the proposition. The idea of the proof is already captured in Example 1 above.

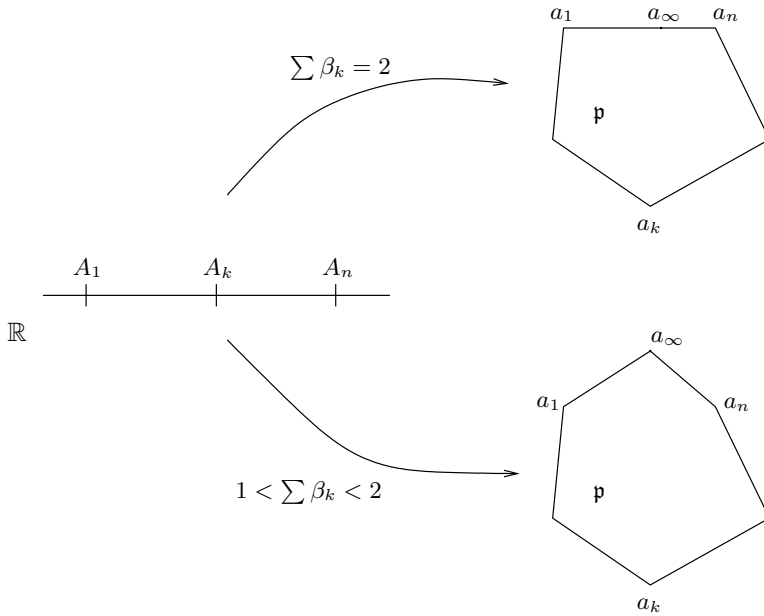


Figure 6. Action of the integral $S(z)$

Proof. We assume that $\sum_{k=1}^n \beta_k = 2$. If $A_k < x < A_{k+1}$ when $1 \leq k \leq n-1$, then

$$S'(x) = \prod_{j \leq k} (x - A_j)^{-\beta_j} \prod_{j > k} (x - A_j)^{-\beta_j}.$$

Hence

$$\arg S'(x) = \arg \left(\prod_{j>k} (x - A_j)^{-\beta_j} \right) = \arg \prod_{j>k} e^{-i\pi\beta_j} = -\pi \sum_{j>k} \beta_j,$$

which of course is constant when x traverses the interval (A_k, A_{k+1}) . Since

$$S(x) = S(A_k) + \int_{A_k}^x S'(y) dy,$$

we see that as x varies from A_k to A_{k+1} , $S(x)$ varies from $S(A_k) = a_k$ to $S(A_{k+1}) = a_{k+1}$ along the straight line segment⁷ $[a_k, a_{k+1}]$, and this makes an angle of $-\pi \sum_{j>k} \beta_j$ with the real axis. Similarly, when $A_n < x$ then $S'(x)$ is positive, while if $x < A_1$, the argument of $S'(x)$ is $-\pi \sum_{k=1}^n \beta_k = -2\pi$, and so $S'(x)$ is again positive. Thus as x varies on $[A_n, +\infty)$, $S(x)$ varies along a straight line (parallel to the x -axis) between a_n and a_∞ ; similarly $S(x)$ varies along a straight line (parallel to that axis) between a_∞ and a_1 as x varies in $(-\infty, A_1]$. Moreover, the union of $[a_n, a_\infty)$ and $(a_\infty, a_1]$ is the segment $[a_n, a_1]$ with the point a_∞ removed.

Now the increase of the angle of $[a_{k+1}, a_k]$ over that of $[a_{k-1}, a_k]$ is $\pi\beta_k$, which means that the angle at the vertex a_k is $\pi\alpha_k$. The proof when $1 < \sum_{k=1}^n \beta_k < 2$ is similar, and is left to the reader.

As elegant as this proposition is, it does not settle the problem of finding a conformal map from the half-plane to a given region P that is bounded by a polygon. There are two reasons for this.

1. It is not true for general n and generic choices of $A_1, \dots, A_n$, that the polygon (which is the image of the real axis under S) is simple, that is, it does not cross itself. Nor is it true in general that the mapping S is conformal on the upper half-plane.
2. Neither does the proposition show that starting with a simply connected region P (whose boundary is a polygonal line $\mathfrak{p}$) the mapping S is, for certain choices of $A_1, \dots, A_n$ and simple modifications, a conformal map from $\mathbb{H}$ to P . That however is the case, and is the result whose proof we now turn to.

⁷We denote the closed straight line segment between two complex numbers z and w by $[z, w]$, that is, $[z, w] = \{(1-t)z + tw : t \in [0, 1]\}$. If we restrict $0 < t < 1$, then (z, w) denotes the open line segment between z and w . Similarly for the half-open segments $[z, w)$ and $(z, w]$ obtained by restricting $0 \leq t < 1$ and $0 < t \leq 1$, respectively.

4.3 Boundary behavior

In what follows we shall consider a **polygonal region** P , namely a bounded, simply connected open set whose boundary is a polygonal line $\mathbf{p}$. In this context, we always assume that the polygonal line is closed, and we sometimes refer to $\mathbf{p}$ as a **polygon**.

To study conformal maps from the half-plane $\mathbb{H}$ to P , we consider first the conformal maps from the disc $\mathbb{D}$ to P , and their boundary behavior.

Theorem 4.2 *If $F : \mathbb{D} \rightarrow P$ is a conformal map, then F extends to a continuous bijection from the closure $\overline{\mathbb{D}}$ of the disc to the closure $\overline{P}$ of the polygonal region. In particular, F gives rise to a bijection from the boundary of the disc to the boundary polygon $\mathbf{p}$.*

The main point consists in showing that if z_0 belongs to the unit circle, then $\lim_{z \rightarrow z_0} F(z)$ exists. To prove this, we need a preliminary result, which uses the fact that if $f : U \rightarrow f(U)$ is conformal, then

$$\text{Area}(f(U)) = \iint_U |f'(z)|^2 dx dy.$$

This assertion follows from the definition, $\text{Area}(f(U)) = \iint_{f(U)} dx dy$, and the fact that the determinant of the Jacobian in the change of variables $w = f(z)$ is simply $|f'(z)|^2$, an observation we made in equation (4), Section 2.2, Chapter 1.

Lemma 4.3 *For each $0 < r < 1/2$, let C_r denote the circle centered at z_0 of radius r . Suppose that for all sufficiently small r we are given two points z_r and z'_r in the unit disc that also lie on C_r . If we let $\rho(r) = |f(z_r) - f(z'_r)|$, then there exists a sequence $\{r_n\}$ of radii that tends to zero, and such that $\lim_{n \rightarrow \infty} \rho(r_n) = 0$.*

Proof. If not, there exist $0 < c$ and $0 < R < 1/2$ such that $c \leq \rho(r)$ for all $0 < r \leq R$. Observe that

$$f(z_r) - f(z'_r) = \int_{\alpha} f'(\zeta) d\zeta,$$

where the integral is taken over the arc α on C_r that joins z_r and z'_r in $\mathbb{D}$. If we parametrize this arc by $z_0 + re^{i\theta}$ with $\theta_1(r) \leq \theta \leq \theta_2(r)$, then

$$\rho(r) \leq \int_{\theta_1(r)}^{\theta_2(r)} |f'(z)| r d\theta.$$

We now apply the Cauchy-Schwarz inequality to see that

$$\rho(r) \leq \left(\int_{\theta_1(r)}^{\theta_2(r)} |f'(z)|^2 r \, d\theta \right)^{1/2} \left(\int_{\theta_1(r)}^{\theta_2(r)} r \, d\theta \right)^{1/2}.$$

Squaring both sides and dividing by r yields

$$\frac{\rho(r)^2}{r} \leq 2\pi \int_{\theta_1(r)}^{\theta_2(r)} |f'(z)|^2 r \, d\theta.$$

We may now integrate both sides from 0 to R , and since $c \leq \rho(r)$ on that region we obtain

$$c^2 \int_0^R \frac{dr}{r} \leq 2\pi \int_0^R \int_{\theta_1(r)}^{\theta_2(r)} |f'(z)|^2 r \, d\theta dr \leq 2\pi \iint_{\mathbb{D}} |f'(z)|^2 \, dx dy.$$

Now the left-hand side is infinite because $1/r$ is not integrable near the origin, and the right-hand side is bounded because the area of the polygonal region is bounded, so this yields the desired contradiction and concludes the proof of the lemma.

Lemma 4.4 *Let z_0 be a point on the unit circle. Then $F(z)$ tends to a limit as z approaches z_0 within the unit disc.*

Proof. If not, there are two sequences $\{z_1, z_2, \dots\}$ and $\{z'_1, z'_2, \dots\}$ in the unit disc that converge to z_0 and are so that $F(z_n)$ and $F(z'_n)$ converge to two distinct points ζ and ζ' in the closure of P . Since F is conformal, the points ζ and ζ' must lie on the boundary $\mathfrak{p}$ of P . We may therefore choose two disjoint discs D and D' centered at ζ and ζ' , respectively, that are at a distance $d > 0$ from each other. For all large n , $F(z_n) \in D$ and $F(z'_n) \in D'$. Therefore, there exist two continuous curves⁸ Λ and Λ' in $D \cap P$ and $D' \cap P$, respectively, with $F(z_n) \in \Lambda$ and $F(z'_n) \in \Lambda'$ for all large n , and with the end-points of Λ and Λ' equal to ζ and ζ' , respectively.

Define $\lambda = F^{-1}(\Lambda)$ and $\lambda' = F^{-1}(\Lambda')$. Then λ and λ' are two continuous curves in $\mathbb{D}$. Moreover, both λ and λ' contain infinitely many points in each sequence $\{z_n\}$ and $\{z'_n\}$. Recall that these sequences converge to z_0 . By continuity, the circle C_r centered at z_0 and of radius r will intersect λ and λ' for all small r , say at some points $z_r \in \lambda$ and $z'_r \in \lambda'$. This

⁸By a continuous curve, we mean the image of a continuous (not necessarily piecewise-smooth) function from a closed interval $[a, b]$ to $\mathbb{C}$.

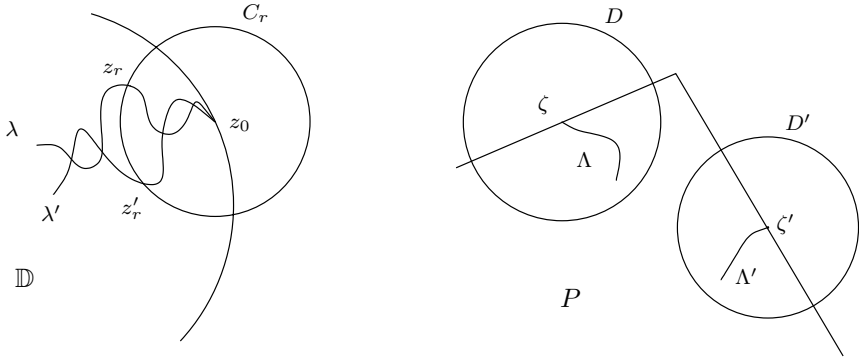


Figure 7. Illustration for the proof of Lemma 4.4

contradicts the previous lemma, because $|F(z_r) - F(z'_r)| > d$. Therefore $F(z)$ converges to a limit on $\mathfrak{p}$ as z approaches z_0 from within the unit disc, and the proof is complete.

Lemma 4.5 *The conformal map F extends to a continuous function from the closure of the disc to the closure of the polygon.*

Proof. By the previous lemma, the limit

$$\lim_{z \rightarrow z_0} F(z)$$

exists, and we define $F(z_0)$ to be the value of this limit. There remains to prove that F is continuous on the closure of the unit disc. Given ϵ , there exists δ such that whenever $z \in \mathbb{D}$ and $|z - z_0| < \delta$, then $|F(z) - F(z_0)| < \epsilon$. Now if z belongs to the boundary of $\mathbb{D}$ and $|z - z_0| < \delta$, then we may choose w such that $|F(z) - F(w)| < \epsilon$ and $|w - z_0| < \delta$. Therefore

$$|F(z) - F(z_0)| \leq |F(z) - F(w)| + |F(w) - F(z_0)| < 2\epsilon,$$

and the lemma is established.

We may now complete the proof of the theorem. We have shown that F extends to a continuous function from $\overline{\mathbb{D}}$ to $\overline{P}$. The previous argument can be applied to the inverse G of F . Indeed, the key geometric property of the unit disc that we used was that if z_0 belongs to the boundary of $\mathbb{D}$, and C is any small circle centered at z_0 , then $C \cap \mathbb{D}$ consists of an arc. Clearly, this property also holds at every boundary point of the

polygonal region P . Therefore, G also extends to a continuous function from $\overline{P}$ to $\overline{\mathbb{D}}$. It suffices to now prove that the extensions of F and G are inverses of each other. If $z \in \partial\mathbb{D}$ and $\{z_k\}$ is a sequence in the disc that converges to z , then $G(F(z_k)) = z_k$, so after taking the limit and using the fact that F is continuous, we conclude that $G(F(z)) = z$ for all $z \in \overline{\mathbb{D}}$. Similarly, $F(G(w)) = w$ for all $w \in \overline{P}$, and the theorem is proved.

The circle of ideas used in this proof can be used to prove more general theorems on the boundary continuity of conformal maps. See Exercise 18 and Problem 6 below.

4.4 The mapping formula

Suppose P is a polygonal region bounded by a polygon $\mathfrak{p}$ whose vertices are ordered consecutively $a_1, a_2, \dots, a_n$, and with $n \geq 3$. We denote by $\pi\alpha_k$ the interior angle at a_k , and define the exterior angle $\pi\beta_k$ by $\alpha_k + \beta_k = 1$. A simple geometric argument provides $\sum_{k=1}^n \beta_k = 2$.

We shall consider conformal mappings of the half-plane $\mathbb{H}$ to P , and make use of the results of the previous section regarding conformal maps from the disc $\mathbb{D}$ to P . The standard correspondences $w = (i - z)/(i + z)$, $z = i(1 - w)/(1 + w)$ allows us to go back and forth between $z \in \mathbb{H}$ and $w \in \mathbb{D}$. Notice that the boundary point $w = -1$ of the circle corresponds to the point at infinity on the line, and so the conformal map of $\mathbb{H}$ to $\mathbb{D}$ extends to a continuous bijection of the boundary of $\mathbb{H}$, which for the purpose of this discussion includes the point at infinity.

Let F be a conformal map from $\mathbb{H}$ to P . (Its existence is guaranteed by the Riemann mapping theorem and the previous discussion.) We assume first that none of the vertices of $\mathfrak{p}$ correspond to the point at infinity. Therefore, there are real numbers $A_1, A_2, \dots, A_n$ so that $F(A_k) = a_k$ for all k . Since F is continuous and injective, and the vertices are numbered consecutively, we may conclude that the A_k 's are in either increasing or decreasing order. After relabeling the vertices a_k and the points A_k , we may assume that $A_1 < A_2 < \dots < A_n$. These points divide the real line into $n - 1$ segments $[A_k, A_{k+1}]$, $1 \leq k \leq n - 1$, and the segment that consists of the join of the two half-segments $(-\infty, A_1] \cup [A_n, \infty)$. These are mapped bijectively onto the corresponding sides of the polygon, that is, the segments $[a_k, a_{k+1}]$, $1 \leq k \leq n - 1$, and $[a_n, a_1]$ (see Figure 8).

Theorem 4.6 *There exist complex numbers c_1 and c_2 so that the conformal map F of $\mathbb{H}$ to P is given by*

$$F(z) = c_1 S(z) + c_2$$

where S is the Schwarz-Christoffel integral introduced in Section 4.2.

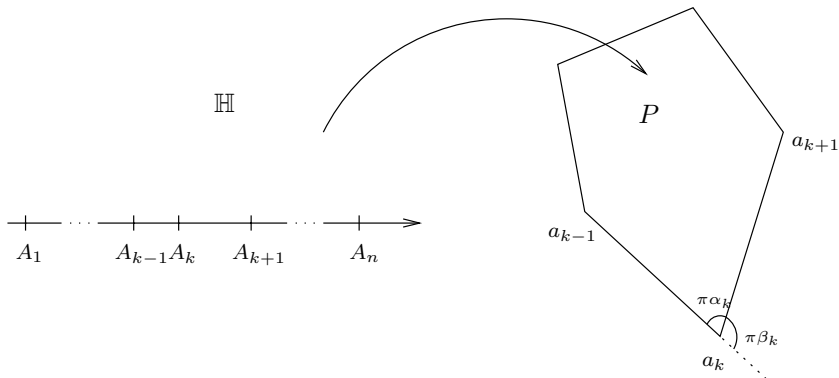


Figure 8. The mapping F

Proof. We first consider z in the upper half-plane lying above the two adjacent segments $[A_{k-1}, A_k]$ and $[A_k, A_{k+1}]$, where $1 < k < n$. We note that F maps these two segments to two segments that intersect at $a_k = F(A_k)$ at an angle $\pi\alpha_k$.

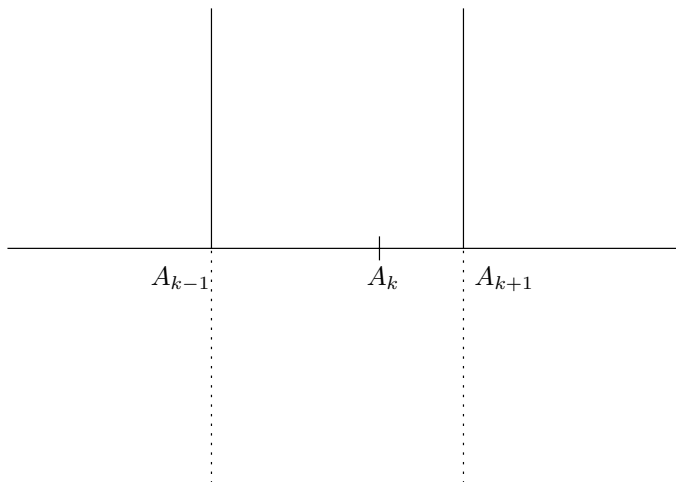
By choosing a branch of the logarithm we can in turn define

$$h_k(z) = (F(z) - a_k)^{1/\alpha_k}$$

for all z in the half-strip in the upper half-plane bounded by the lines $\operatorname{Re}(z) = A_{k-1}$ and $\operatorname{Re}(z) = A_{k+1}$. Since F continues to the boundary of $\mathbb{H}$, the map h_k is actually continuous up to the segment (A_{k-1}, A_{k+1}) on the real line. By construction h_k will map the segment $[A_{k-1}, A_{k+1}]$ to a (straight) segment L_k in the complex plane, with A_k mapped to 0. We may therefore apply the Schwarz reflection principle to see that h_k is analytically continuable to a holomorphic function in the two-way infinite strip $A_{k-1} < \operatorname{Re}(z) < A_{k+1}$ (see Figure 9). We claim that h'_k never vanishes in that strip. First, if z belongs to the open upper half-strip, then

$$\frac{F'(z)}{F(z) - F(A_k)} = \alpha_k \frac{h'_k(z)}{h_k(z)},$$

and since F is conformal, we have $F'(z) \neq 0$ so $h'_k(z) \neq 0$ (Proposition 1.1). By reflection, this also holds in the lower half-strip, and it remains to investigate points on the segment (A_{k-1}, A_{k+1}) . If $A_{k-1} < x < A_{k+1}$, we note that the image under h_k of a small half-disc centered at x and contained in $\mathbb{H}$ lies on one side of the straight line segment

**Figure 9.** Schwarz reflection

L_k . Since h_k is injective up to L_k (because F is) the symmetry in the Schwarz reflection principle guarantees that h_k is injective in the whole disc centered at x , whence $h'_k(x) \neq 0$, whence $h'_k(z) \neq 0$ for all z in the strip $A_{k-1} < \operatorname{Re}(z) < A_{k+1}$.

Now because $F' = \alpha_k h_k^{-\beta_k} h'_k$ and $F'' = -\beta_k \alpha_k h_k^{-\beta_k-1} (h'_k)^2 + \alpha_k h_k^{-\beta_k} h''_k$, the fact that $h'_k(z) \neq 0$ implies that

$$\frac{F''(z)}{F'(z)} = \frac{-\beta_k}{z - A_k} + E_k(z),$$

where E_k is holomorphic in the strip $A_{k-1} < \operatorname{Re}(z) < A_{k+1}$. A similar result holds for $k = 1$ and $k = n$, namely

$$\frac{F''(z)}{F'(z)} = -\frac{\beta_1}{z - A_1} + E_1,$$

where E_1 is holomorphic in the strip $-\infty < \operatorname{Re}(z) < A_2$, and

$$\frac{F''(z)}{F'(z)} = -\frac{\beta_n}{z - A_n} + E_n,$$

where E_n is holomorphic in the strip $A_{n-1} < \operatorname{Re}(z) < \infty$. Finally, another application of the reflection principle shows that F is continuable in the exterior of a disc $|z| \leq R$, for large R (say $R > \max_{1 \leq k \leq n} |A_k|$). Indeed, we may continue F across the union of the segments

$(-\infty, A_1) \cup (A_n, \infty)$ since their image under F is a straight line segment and Schwarz reflection applies. The fact that F maps the upper half-plane to a bounded region shows that the analytic continuation of F outside a large disc is also bounded, and hence holomorphic at infinity. Thus F''/F' is holomorphic at infinity and we claim that it goes to 0 as $|z| \rightarrow \infty$. Indeed, we may expand F at $z = \infty$ as

$$F(z) = c_0 + \frac{c_1}{z} + \frac{c_2}{z^2} + \cdots.$$

This after differentiation shows that F''/F' decays like $1/z$ as $|z|$ becomes large, and proves our claim.

Altogether then, because the various strips overlap and cover the entire complex plane,

$$\frac{F''(z)}{F'(z)} + \sum_{k=1}^n \frac{\beta_k}{z - A_k}$$

is holomorphic in the entire plane and vanishes at infinity; thus, by Liouville's theorem it is zero. Hence

$$\frac{F''(z)}{F'(z)} = - \sum_{k=1}^n \frac{\beta_k}{z - A_k}.$$

From this we contend that $F'(z) = c(z - A_1)^{-\beta_1} \cdots (z - A_n)^{-\beta_n}$. Indeed, denoting this product by $Q(z)$, we have

$$\frac{Q'(z)}{Q(z)} = - \sum_{k=1}^n \frac{\beta_k}{z - A_k}.$$

Therefore

$$\frac{d}{dz} \left(\frac{F'(z)}{Q(z)} \right) = 0,$$

which proves the contention. A final integration yields the theorem.

We may now withdraw the hypothesis we made at the beginning that F did not map the point at infinity to a vertex of P , and obtain a formula for that case as well.

Theorem 4.7 *If F is a conformal map from the upper half-plane to the polygonal region P and maps the points $A_1, \dots, A_{n-1}, \infty$ to the vertices of $\mathfrak{p}$, then there exist constants C_1 and C_2 such that*

$$F(z) = C_1 \int_0^z \frac{d\zeta}{(\zeta - A_1)^{\beta_1} \cdots (\zeta - A_{n-1})^{\beta_{n-1}}} + C_2.$$

In other words, the formula is obtained by deleting the last term in the Schwarz-Christoffel integral (5).

Proof. After a preliminary translation, we may assume that $A_j \neq 0$ for $j = 1, \dots, n-1$. Choose a point $A_n^* > 0$ on the real line, and consider the fractional linear map defined by

$$\Phi(z) = A_n^* - \frac{1}{z}.$$

Then Φ is an automorphism of the upper half-plane. Let $A_k^* = \Phi(A_k)$ for $k = 1, \dots, n-1$, and note that $A_n^* = \Phi(\infty)$. Then

$$(F \circ \Phi^{-1})(A_k^*) = a_k \quad \text{for all } k = 1, 2, \dots, n.$$

We can now apply the Schwarz-Christoffel formula just proved to find that

$$(F \circ \Phi^{-1})(z') = C_1 \int_0^{z'} \frac{d\zeta}{(\zeta - A_1^*)^{\beta_1} \dots (\zeta - A_n^*)^{\beta_n}} + C_2.$$

The change of variables $\zeta = \Phi(w)$ satisfies $d\zeta = dw/w^2$, and since we can write $2 = \beta_1 + \dots + \beta_n$, we obtain

$$\begin{aligned} (F \circ \Phi^{-1})(z') &= C_1 \int_0^{\Phi^{-1}(z')} \frac{dw}{(w(A_n^* - A_1^*) - 1)^{\beta_1} \dots (w(A_n^* - A_{n-1}^*) - 1)^{\beta_{n-1}}} + C_2' \\ &= C_1' \int_0^{\Phi^{-1}(z')} \frac{dw}{(w - 1/(A_n^* - A_1^*))^{\beta_1} \dots (w - 1/(A_n^* - A_{n-1}^*))^{\beta_{n-1}}} + C_2'. \end{aligned}$$

Finally, we note that $1/(A_n^* - A_k^*) = A_k$ and set $\Phi^{-1}(z') = z$ in the above equation to conclude that

$$F(z) = C_1' \int_0^z \frac{dw}{(w - A_1)^{\beta_1} \dots (w - A_{n-1})^{\beta_{n-1}}} + C_2',$$

as was to be shown.

4.5 Return to elliptic integrals

We consider again the elliptic integral

$$I(z) = \int_0^z \frac{d\zeta}{[(1 - \zeta^2)(1 - k^2\zeta^2)]^{1/2}} \quad \text{with } 0 < k < 1,$$

which arose in Example 3 of Section 4.1. We saw that it mapped the real axis to the rectangle R with vertices $-K$, K , $K + iK'$, and $-K + iK'$. We will now see that this mapping is a conformal mapping of $\mathbb{H}$ to the interior of R .

According to Theorem 4.6 there is a conformal map F to the rectangle, that maps four points on the real axis to the vertices of R . By preceding this map with a suitable automorphism of $\mathbb{H}$ we may assume that F maps -1 , 0 , 1 to $-K$, 0 , K , respectively. Indeed, by using a preliminary automorphism, we may assume that $-K$, 0 , K are the images of points A_1 , 0 , A_2 with $A_1 < 0 < A_2$; then we can further take $A_1 = -1$ and $A_2 = 1$. See Exercise 15.

Next, let ℓ be chosen with $0 < \ell < 1$, so that $1/\ell$ is the point on the real line mapped by F to the vertex $K + iK'$, which is the vertex next in order after $-K$ and K . We claim that $F(-1/\ell)$ is the vertex $-K + iK'$. Indeed, if $F^*(z) = -F(-\bar{z})$, then by the symmetry of R , F^* is also a conformal map of $\mathbb{H}$ to R ; moreover $F^*(0) = 0$, and $F^*(\pm 1) = \pm K$. Thus $F^{-1} \circ F^*$ is an automorphism of $\mathbb{H}$ that fixes the points -1 , 0 , and 1 . Hence $F^{-1} \circ F^*$ is the identity (see Exercise 15), and $F = F^*$, from which it follows that

$$F(-1/\ell) = -\overline{F}(1/\ell) = -K + iK'.$$

Therefore, by Theorem 4.6

$$F(z) = c_1 \int_0^z \frac{d\zeta}{[(1 - \zeta^2)(1 - \ell^2 \zeta^2)]^{1/2}} + c_2.$$

Setting $z = 0$ gives $c_2 = 0$, and letting $z = 1$, $z = 1/\ell$, yields

$$K(k) = c_1 K(\ell) \quad \text{and} \quad K'(k) = c_1 K'(\ell),$$

where

$$K(k) = \int_0^1 \frac{dx}{[(1 - x^2)(1 - k^2 x^2)]^{1/2}},$$

$$K'(k) = \int_1^{1/k} \frac{dx}{[(x^2 - 1)(1 - k^2 x^2)]^{1/2}}.$$

Now $K(k)$ is clearly strictly increasing as k varies in $(0, 1)$. Moreover, a change of variables (Exercise 24) establishes the identity

$$K'(k) = K(\tilde{k}) \quad \text{where } \tilde{k}^2 = 1 - k^2 \text{ and } \tilde{k} > 0,$$

and this shows that $K'(k)$ is strictly decreasing. Hence $K(k)/K'(k)$ is strictly increasing. Since $K(k)/K'(k) = K(\ell)/K'(\ell)$, we must have $k = \ell$, and finally $c_1 = 1$. This shows that $I(z) = F(z)$, and hence I is conformal, as was to be proved.

A final observation is of significance. A basic insight into elliptic integrals is obtained by passing to their inverse functions. We therefore consider $z \mapsto \operatorname{sn}(z)$, the inverse map of $z \mapsto I(z)$.⁹ It transforms the closed rectangle into the closed upper half-plane. Now consider the series of rectangles $R = R_0, R_1, R_2, \dots$ gotten by reflecting successively along the lower sides (Figure 10).

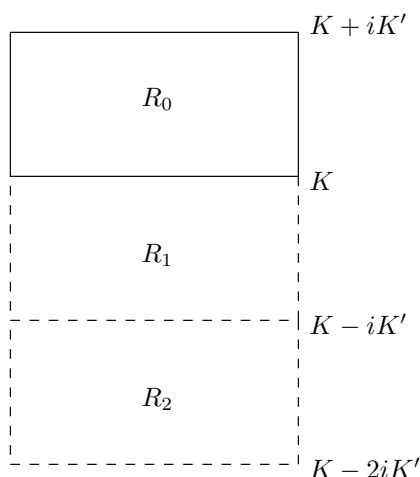


Figure 10. Reflections of $R = R_0$

With $\operatorname{sn}(z)$ defined in R_0 , we can by the reflection principle extend it to R_1 by setting $\operatorname{sn}(z) = \overline{\operatorname{sn}(\bar{z})}$ whenever $z \in R_1$ (note that then $\bar{z} \in R_0$). Next we can extend $\operatorname{sn}(z)$ to R_2 by setting $\operatorname{sn}(z) = \overline{\operatorname{sn}(-iK' + \bar{z})}$ if $z \in R_2$ and noting that if $z \in R_2$, then $-iK' + \bar{z} \in R_1$. Combining these reflections and continuing this way we see that we can extend $\operatorname{sn}(z)$ in the entire strip $-K < \operatorname{Re}(z) < K$, so that $\operatorname{sn}(z) = \operatorname{sn}(z + 2iK')$.

Similarly, by reflecting in a series of horizontal rectangles, and combining these with the previous reflections, we see that $\operatorname{sn}(z)$ can be continued to the complex plane and also satisfies $\operatorname{sn}(z) = \operatorname{sn}(z + 4K)$. Thus $\operatorname{sn}(z)$ is *doubly periodic* (with periods $4K$ and $2iK'$). A further examination

⁹The notation $\operatorname{sn}(z)$ in somewhat different form is due to Jacobi, and was adopted because of the analogy with $\sin z$.

shows that the only singularities $\text{sn}(z)$ are poles. Functions of this type, called “elliptic functions,” are the subject of the next chapter.

5 Exercises

1. A holomorphic mapping $f : U \rightarrow V$ is a **local bijection** on U if for every $z \in U$ there exists an open disc $D \subset U$ centered at z , so that $f : D \rightarrow f(D)$ is a bijection.

Prove that a holomorphic map $f : U \rightarrow V$ is a local bijection on U if and only if $f'(z) \neq 0$ for all $z \in U$.

[Hint: Use Rouché’s theorem as in the proof of Proposition 1.1.]

2. Suppose $F(z)$ is holomorphic near $z = z_0$ and $F(z_0) = F'(z_0) = 0$, while $F''(z_0) \neq 0$. Show that there are two curves Γ_1 and Γ_2 that pass through z_0 , are orthogonal at z_0 , and so that F restricted to Γ_1 is real and has a minimum at z_0 , while F restricted to Γ_2 is also real but has a maximum at z_0 .

[Hint: Write $F(z) = (g(z))^2$ for z near z_0 , and consider the mapping $z \mapsto g(z)$ and its inverse.]

3. Suppose U and V are conformally equivalent. Prove that if U is simply connected, then so is V . Note that this conclusion remains valid if we merely assume that there exists a continuous bijection between U and V .

4. Does there exist a holomorphic surjection from the unit disc to $\mathbb{C}$?

[Hint: Move the upper half-plane “down” and then square it to get $\mathbb{C}$.]

5. Prove that $f(z) = -\frac{1}{2}(z + 1/z)$ is a conformal map from the half-disc $\{z = x + iy : |z| < 1, y > 0\}$ to the upper half-plane.

[Hint: The equation $f(z) = w$ reduces to the quadratic equation $z^2 + 2wz + 1 = 0$, which has two distinct roots in $\mathbb{C}$ whenever $w \neq \pm 1$. This is certainly the case if $w \in \mathbb{H}$.]

6. Give another proof of Lemma 1.3 by showing directly that the Laplacian of $u \circ F$ is zero.

[Hint: The real and imaginary parts of F satisfy the Cauchy-Riemann equations.]

7. Provide all the details in the proof of the formula for the solution of the Dirichlet problem in a strip discussed in Section 1.3. Recall that it suffices to compute the solution at the points $z = iy$ with $0 < y < 1$.

(a) Show that if $re^{i\theta} = G(iy)$, then

$$re^{i\theta} = i \frac{\cos \pi y}{1 + \sin \pi y}.$$

This leads to two separate cases: either $0 < y \leq 1/2$ and $\theta = \pi/2$, or $1/2 \leq$

$y < 1$ and $\theta = -\pi/2$. In either case, show that

$$r^2 = \frac{1 - \sin \pi y}{1 + \sin \pi y} \quad \text{and} \quad P_r(\theta - \varphi) = \frac{\sin \pi y}{1 - \cos \pi y \sin \varphi}.$$

- (b) In the integral $\frac{1}{2\pi} \int_0^\pi P_r(\theta - \varphi) \tilde{f}_0(\varphi) d\varphi$ make the change of variables $t = F(e^{i\varphi})$. Observe that

$$e^{i\varphi} = \frac{i - e^{\pi t}}{i + e^{\pi t}},$$

and then take the imaginary part and differentiate both sides to establish the two identities

$$\sin \varphi = \frac{1}{\cosh \pi t} \quad \text{and} \quad \frac{d\varphi}{dt} = \frac{\pi}{\cosh \pi t}.$$

Hence deduce that

$$\begin{aligned} \frac{1}{2\pi} \int_0^\pi P_r(\theta - \varphi) \tilde{f}_0(\varphi) d\varphi &= \frac{1}{2\pi} \int_0^\pi \frac{\sin \pi y}{1 - \cos \pi y \sin \varphi} \tilde{f}_0(\varphi) d\varphi \\ &= \frac{\sin \pi y}{2} \int_{-\infty}^\infty \frac{f_0(t)}{\cosh \pi t - \cos \pi y} dt. \end{aligned}$$

- (c) Use a similar argument to prove the formula for the integral $\frac{1}{2\pi} \int_{-\pi}^0 P_r(\theta - \varphi) \tilde{f}_1(\varphi) d\varphi$.

8. Find a harmonic function u in the open first quadrant that extends continuously up to the boundary except at the points 0 and 1, and that takes on the following boundary values: $u(x, y) = 1$ on the half-lines $\{y = 0, x > 1\}$ and $\{x = 0, y > 0\}$, and $u(x, y) = 0$ on the segment $\{0 < x < 1, y = 0\}$.

[Hint: Find conformal maps $F_1, F_2, \dots, F_5$ indicated in Figure 11. Note that $\frac{1}{\pi} \arg(z)$ is harmonic on the upper half-plane, equals 0 on the positive real axis, and 1 on the negative real axis.]

9. Prove that the function u defined by

$$u(x, y) = \operatorname{Re} \left(\frac{i + z}{i - z} \right) \quad \text{and} \quad u(0, 1) = 0$$

is harmonic in the unit disc and vanishes on its boundary. Note that u is not bounded in $\mathbb{D}$.

10. Let $F: \mathbb{H} \rightarrow \mathbb{C}$ be a holomorphic function that satisfies

$$|F(z)| \leq 1 \quad \text{and} \quad F(i) = 0.$$

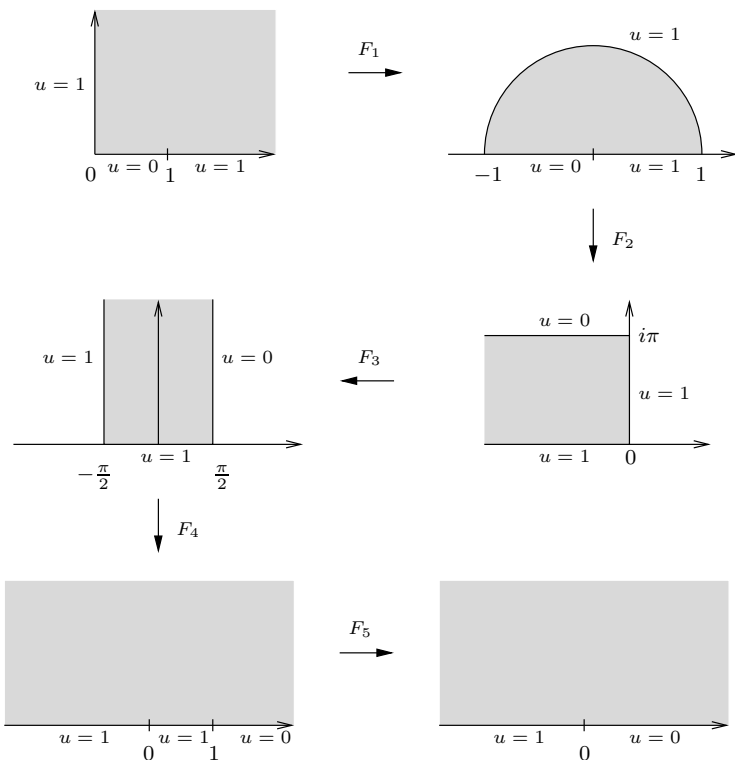


Figure 11. Successive conformal maps in Exercise 8

Prove that

$$|F(z)| \leq \left| \frac{z-i}{z+i} \right| \quad \text{for all } z \in \mathbb{H}.$$

11. Show that if $f : D(0, R) \rightarrow \mathbb{C}$ is holomorphic, with $|f(z)| \leq M$ for some $M > 0$, then

$$\left| \frac{f(z) - f(0)}{M^2 - \overline{f(0)}f(z)} \right| \leq \frac{|z|}{MR}.$$

[Hint: Use the Schwarz lemma.]

12. A complex number $w \in \mathbb{D}$ is a **fixed point** for the map $f : \mathbb{D} \rightarrow \mathbb{D}$ if $f(w) = w$.

- (a) Prove that if $f : \mathbb{D} \rightarrow \mathbb{D}$ is analytic and has two distinct fixed points, then f is the identity, that is, $f(z) = z$ for all $z \in \mathbb{D}$.

- (b) Must every holomorphic function $f : \mathbb{D} \rightarrow \mathbb{D}$ have a fixed point? [Hint: Consider the upper half-plane.]

13. The **pseudo-hyperbolic distance** between two points $z, w \in \mathbb{D}$ is defined by

$$\rho(z, w) = \left| \frac{z - w}{1 - \overline{w}z} \right|.$$

- (a) Prove that if $f : \mathbb{D} \rightarrow \mathbb{D}$ is holomorphic, then

$$\rho(f(z), f(w)) \leq \rho(z, w) \quad \text{for all } z, w \in \mathbb{D}.$$

Moreover, prove that if f is an automorphism of $\mathbb{D}$ then f preserves the pseudo-hyperbolic distance

$$\rho(f(z), f(w)) = \rho(z, w) \quad \text{for all } z, w \in \mathbb{D}.$$

[Hint: Consider the automorphism $\psi_\alpha(z) = (z - \alpha)/(1 - \overline{\alpha}z)$ and apply the Schwarz lemma to $\psi_{f(w)} \circ f \circ \psi_w^{-1}$.]

- (b) Prove that

$$\frac{|f'(z)|}{1 - |f(z)|^2} \leq \frac{1}{1 - |z|^2} \quad \text{for all } z \in \mathbb{D}.$$

This result is called the Schwarz-Pick lemma. See Problem 3 for an important application of this lemma.

14. Prove that all conformal mappings from the upper half-plane $\mathbb{H}$ to the unit disc $\mathbb{D}$ take the form

$$e^{i\theta} \frac{z - \beta}{z - \overline{\beta}}, \quad \theta \in \mathbb{R} \text{ and } \beta \in \mathbb{H}.$$

15. Here are two properties enjoyed by automorphisms of the upper half-plane.

- (a) Suppose Φ is an automorphism of $\mathbb{H}$ that fixes three distinct points on the real axis. Then Φ is the identity.
- (b) Suppose (x_1, x_2, x_3) and (y_1, y_2, y_3) are two pairs of three distinct points on the real axis with

$$x_1 < x_2 < x_3 \quad \text{and} \quad y_1 < y_2 < y_3.$$

Prove that there exists (a unique) automorphism Φ of $\mathbb{H}$ so that $\Phi(x_j) = y_j$, $j = 1, 2, 3$. The same conclusion holds if $y_3 < y_1 < y_2$ or $y_2 < y_3 < y_1$.

16. Let

$$f(z) = \frac{i-z}{i+z} \quad \text{and} \quad f^{-1}(w) = i \frac{1-w}{1+w}.$$

- (a) Given $\theta \in \mathbb{R}$, find real numbers a, b, c, d such that $ad - bc = 1$, and so that for any $z \in \mathbb{H}$

$$\frac{az+b}{cz+d} = f^{-1} \left(e^{i\theta} f(z) \right).$$

- (b) Given $\alpha \in \mathbb{D}$ find real numbers a, b, c, d so that $ad - bc = 1$, and so that for any $z \in \mathbb{H}$

$$\frac{az+b}{cz+d} = f^{-1}(\psi_\alpha(f(z))),$$

with ψ_α defined in Section 2.1.

- (c) Prove that if g is an automorphism of the unit disc, then there exist real numbers a, b, c, d such that $ad - bc = 1$ and so that for any $z \in \mathbb{H}$

$$\frac{az+b}{cz+d} = f^{-1} \circ g \circ f(z).$$

[Hint: Use parts (a) and (b).]

17. If $\psi_\alpha(z) = (\alpha - z)/(1 - \bar{\alpha}z)$ for $|\alpha| < 1$, prove that

$$\frac{1}{\pi} \iint_{\mathbb{D}} |\psi'_\alpha|^2 dx dy = 1 \quad \text{and} \quad \frac{1}{\pi} \iint_{\mathbb{D}} |\psi'_\alpha| dx dy = \frac{1 - |\alpha|^2}{|\alpha|^2} \log \frac{1}{1 - |\alpha|^2},$$

where in the case $\alpha = 0$ the expression on the right is understood as the limit as $|\alpha| \rightarrow 0$.

[Hint: The first integral can be evaluated without a calculation. For the second, use polar coordinates, and for each fixed r use contour integration to evaluate the integral in θ .]

18. Suppose that Ω is a simply connected domain that is bounded by a piecewise-smooth closed curve γ (in the terminology of Chapter 1). Then any conformal map F of $\mathbb{D}$ to Ω extends to a continuous bijection of $\bar{\mathbb{D}}$ to $\bar{\Omega}$. The proof is simply a generalization of the argument used in Theorem 4.2.

19. Prove that the complex plane slit along the union of the rays $\cup_{k=1}^n \{A_k + iy : y \leq 0\}$ is simply connected.

[Hint: Given a curve, first “raise” it so that it is completely contained in the upper half-plane.]

20. Other examples of elliptic integrals providing conformal maps from the upper half-plane to rectangles are given below.

- (a) The function

$$\int_0^z \frac{d\zeta}{\sqrt{\zeta(\zeta-1)(\zeta-\lambda)}}, \quad \text{with } \lambda \in \mathbb{R} \text{ and } \lambda \neq 1$$

maps the upper half-plane conformally to a rectangle, one of whose vertices is the image of the point at infinity.

- (b) In the case
- $\lambda = -1$
- , the image of

$$\int_0^z \frac{d\zeta}{\sqrt{\zeta(\zeta^2-1)}}$$

is a square whose side lengths are $\frac{\Gamma^2(1/4)}{2\sqrt{2}\pi}$.

21. We consider conformal mappings to triangles.

- (a) Show that

$$\int_0^z z^{-\beta_1}(1-z)^{-\beta_2} dz,$$

with $0 < \beta_1 < 1$, $0 < \beta_2 < 1$, and $1 < \beta_1 + \beta_2 < 2$, maps $\mathbb{H}$ to a triangle whose vertices are the images of 0, 1, and ∞ , and with angles $\alpha_1\pi$, $\alpha_2\pi$, and $\alpha_3\pi$, where $\alpha_j + \beta_j = 1$ and $\beta_1 + \beta_2 + \beta_3 = 2$.

- (b) What happens when $\beta_1 + \beta_2 = 1$?
- (c) What happens when $0 < \beta_1 + \beta_2 < 1$?
- (d) In (a), the length of the side of the triangle opposite angle $\alpha_j\pi$ is $\frac{\sin(\alpha_j\pi)}{\pi}\Gamma(\alpha_1)\Gamma(\alpha_2)\Gamma(\alpha_3)$.

22. If P is a simply connected region bounded by a polygon with vertices $a_1, \dots, a_n$ and angles $\alpha_1\pi, \dots, \alpha_n\pi$, and F is a conformal map of the disc $\mathbb{D}$ to P , then there exist complex numbers $B_1, \dots, B_n$ on the unit circle, and constants c_1 and c_2 so that

$$F(z) = c_1 \int_1^z \frac{d\zeta}{(\zeta - B_1)^{\beta_1} \dots (\zeta - B_n)^{\beta_n}} + c_2.$$

[Hint: This follows from the standard correspondence between $\mathbb{H}$ and $\mathbb{D}$ and an argument similar to that used in the proof of Theorem 4.7.]

23. If

$$F(z) = \int_1^z \frac{d\zeta}{(1-\zeta^n)^{2/n}},$$

then F maps the unit disc conformally onto the interior of a regular polygon with n sides and perimeter

$$2^{\frac{n-2}{n}} \int_0^\pi (\sin \theta)^{-2/n} d\theta.$$

24. The elliptic integrals K and K' defined for $0 < k < 1$ by

$$K(k) = \int_0^1 \frac{dx}{((1-x^2)(1-k^2x^2))^{1/2}} \quad \text{and} \quad K'(k) = \int_1^{1/k} \frac{dx}{((x^2-1)(1-k^2x^2))^{1/2}}$$

satisfy various interesting identities. For instance:

- (a) Show that if $\tilde{k}^2 = 1 - k^2$ and $0 < \tilde{k} < 1$, then

$$K'(\tilde{k}) = K(k).$$

[Hint: Change variables $x = (1 - \tilde{k}^2 y^2)^{-1/2}$ in the integral defining $K'(\tilde{k})$.]

- (b) Prove that if $\tilde{k}^2 = 1 - k^2$, and $0 < \tilde{k} < 1$, then

$$K(k) = \frac{2}{1 + \tilde{k}} K\left(\frac{1 - \tilde{k}}{1 + \tilde{k}}\right).$$

[Hint: Change variables $x = 2t/(1 + \tilde{k} + (1 - \tilde{k})t^2)$.]

- (c) Show that for $0 < k < 1$ one has

$$K(k) = \frac{\pi}{2} F(1/2, 1/2, 1; k^2),$$

where F the hypergeometric series. [Hint: This follows from the integral representation for F given in Exercise 9, Chapter 6.]

6 Problems

1. Let f be a complex-valued C^1 function defined in the neighborhood of a point z_0 . There are several notions closely related to conformality at z_0 . We say that f is **isogonal** at z_0 if whenever $\gamma(t)$ and $\eta(t)$ are two smooth curves with $\gamma(0) = \eta(0) = z_0$, that make an angle θ there ($|\theta| < \pi$), then $f(\gamma(t))$ and $f(\eta(t))$ make an angle of θ' at $t = 0$ with $|\theta'| = |\theta|$ for all θ . Also, f is said to be **isotropic** if it magnifies lengths by some factor for all directions emanating from z_0 , that is, if the limit

$$\lim_{r \rightarrow 0} \frac{|f(z_0 + re^{i\theta}) - f(z_0)|}{r}$$

exists, is non-zero, and independent of θ .

Then f is isogonal at z_0 if and only if it is isotropic at z_0 ; moreover, f is isogonal at z_0 if and only if either $f'(z_0)$ exists and is non-zero, or the same holds for f replaced by $\bar{f}$.

2. The angle between two non-zero complex numbers z and w (taken in that order) is simply the oriented angle, in $(-\pi, \pi]$, that is formed between the two vectors in $\mathbb{R}^2$ corresponding to the points z and w . This oriented angle, say α , is uniquely determined by the two quantities

$$\frac{(z, w)}{|z||w|} \quad \text{and} \quad \frac{(z, -iw)}{|z||w|}$$

which are simply the cosine and sine of α , respectively. Here, the notation $(\cdot, \cdot)$ corresponds to the usual Euclidian inner product in $\mathbb{R}^2$, which in terms of complex numbers takes the form $(z, w) = \operatorname{Re}(z\bar{w})$.

In particular, we may now consider two smooth curves $\gamma : [a, b] \rightarrow \mathbb{C}$ and $\eta : [a, b] \rightarrow \mathbb{C}$, that intersect at z_0 , say $\gamma(t_0) = \eta(t_0) = z_0$, for some $t_0 \in (a, b)$. If the quantities $\gamma'(t_0)$ and $\eta'(t_0)$ are non-zero, then they represent the tangents to the curves γ and η at the point z_0 , and we say that the two curves intersect at z_0 at the angle formed by the two vectors $\gamma'(t_0)$ and $\eta'(t_0)$.

A holomorphic function f defined near z_0 is said to **preserve angles** at z_0 if for any two smooth curves γ and η intersecting at z_0 , the angle formed between the curves γ and η at z_0 equals the angle formed between the curves $f \circ \gamma$ and $f \circ \eta$ at $f(z_0)$. (See Figure 12 for an illustration.) In particular, we assume that the tangents to the curves γ , η , $f \circ \gamma$, and $f \circ \eta$ at the point z_0 and $f(z_0)$ are all non-zero.



Figure 12. Preservation of angles at z_0

- (a) Prove that if $f : \Omega \rightarrow \mathbb{C}$ is holomorphic, and $f'(z_0) \neq 0$, then f preserves angles at z_0 . [Hint: Observe that

$$(f'(z_0)\gamma'(t_0), f'(z_0)\eta'(t_0)) = |f'(z_0)|^2(\gamma'(t_0), \eta'(t_0)).]$$

- (b) Conversely, prove the following: suppose $f : \Omega \rightarrow \mathbb{C}$ is a complex-valued function, that is real-differentiable at $z_0 \in \Omega$, and $J_f(z_0) \neq 0$. If f preserves angles at z_0 , then f is holomorphic at z_0 with $f'(z_0) \neq 0$.

3.* The Schwarz-Pick lemma (see Exercise 13) is the infinitesimal version of an important observation in complex analysis and geometry.

For complex numbers $w \in \mathbb{C}$ and $z \in \mathbb{D}$ we define the **hyperbolic length** of w at z by

$$\|w\|_z = \frac{|w|}{1 - |z|^2},$$

where $|w|$ and $|z|$ denote the usual absolute values. This length is sometimes referred to as the **Poincaré metric**, and as a Riemann metric it is written as

$$ds^2 = \frac{|dz|^2}{(1 - |z|^2)^2}.$$

The idea is to think of w as a vector lying in the tangent space at z . Observe that for a fixed w , its hyperbolic length grows to infinity as z approaches the boundary of the disc. We pass from the infinitesimal hyperbolic length of tangent vectors to the global hyperbolic distance between two points by integration.

- (a) Given two complex numbers z_1 and z_2 in the disc, we define the **hyperbolic distance** between them by

$$d(z_1, z_2) = \inf_{\gamma} \int_0^1 \|\gamma'(t)\|_{\gamma(t)} dt,$$

where the infimum is taken over all smooth curves $\gamma : [0, 1] \rightarrow \mathbb{D}$ joining z_1 and z_2 . Use the Schwarz-Pick lemma to prove that if $f : \mathbb{D} \rightarrow \mathbb{D}$ is holomorphic, then

$$d(f(z_1), f(z_2)) \leq d(z_1, z_2) \quad \text{for any } z_1, z_2 \in \mathbb{D}.$$

In other words, holomorphic functions are distance-decreasing in the hyperbolic metric.

- (b) Prove that automorphisms of the unit disc preserve the hyperbolic distance, namely

$$d(\varphi(z_1), \varphi(z_2)) = d(z_1, z_2), \quad \text{for any } z_1, z_2 \in \mathbb{D}$$

and any automorphism φ . Conversely, if $\varphi : \mathbb{D} \rightarrow \mathbb{D}$ preserves the hyperbolic distance, then either φ or $\overline{\varphi}$ is an automorphism of $\mathbb{D}$.

- (c) Given two points $z_1, z_2 \in \mathbb{D}$, show that there exists an automorphism φ such that $\varphi(z_1) = 0$ and $\varphi(z_2) = s$ for some s on the segment $[0, 1]$ on the real line.
- (d) Prove that the hyperbolic distance between 0 and $s \in [0, 1]$ is

$$d(0, s) = \frac{1}{2} \log \frac{1+s}{1-s}.$$

- (e) Find a formula for the hyperbolic distance between any two points in the unit disc.

4.* Consider the group of matrices of the form

$$M = \begin{pmatrix} a & b \\ c & d \end{pmatrix},$$

that satisfy the following conditions:

- (i) a, b, c , and $d \in \mathbb{C}$,
- (ii) the determinant of M is equal to 1,
- (iii) the matrix M preserves the following hermitian form on $\mathbb{C}^2 \times \mathbb{C}^2$:

$$\langle Z, W \rangle = z_1 \bar{w}_1 - z_2 \bar{w}_2,$$

where $Z = (z_1, z_2)$ and $W = (w_1, w_2)$. In other words, for all $Z, W \in \mathbb{C}^2$

$$\langle MZ, MW \rangle = \langle Z, W \rangle.$$

This group of matrices is denoted by $SU(1, 1)$.

- (a) Prove that all matrices in $SU(1, 1)$ are of the form

$$\begin{pmatrix} a & b \\ \bar{b} & \bar{a} \end{pmatrix},$$

where $|a|^2 - |b|^2 = 1$. To do so, consider the matrix

$$J = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix},$$

and observe that $\langle Z, W \rangle = {}^t W J Z$, where ${}^t W$ denotes the conjugate transpose of W .

- (b) To every matrix in $SU(1, 1)$ we can associate a fractional linear transformation

$$\frac{az + b}{cz + d}.$$

Prove that the group $SU(1, 1)/\{\pm 1\}$ is isomorphic to the group of automorphisms of the disc. [Hint: Use the following association.]

$$e^{2i\theta} \frac{z - \alpha}{1 - \bar{\alpha}z} \longrightarrow \begin{pmatrix} \frac{e^{i\theta}}{\sqrt{1-|\alpha|^2}} & -\frac{\alpha e^{i\theta}}{\sqrt{1-|\alpha|^2}} \\ -\frac{\bar{\alpha} e^{-i\theta}}{\sqrt{1-|\alpha|^2}} & \frac{e^{-i\theta}}{\sqrt{1-|\alpha|^2}} \end{pmatrix}.$$

5. The following result is relevant to Problem 4 in Chapter 10 which treats modular functions.

- (a) Suppose that $F : \mathbb{H} \rightarrow \mathbb{C}$ is holomorphic and bounded. Also, suppose that $F(z)$ vanishes when $z = ir_n$, $n = 1, 2, 3, \dots$, where $\{r_n\}$ is a bounded sequence of positive numbers. Prove that if $\sum_{n=1}^{\infty} r_n = \infty$, then $F = 0$.
- (b) If $\sum r_n < \infty$, it is possible to construct a bounded function on the upper half-plane with zeros precisely at the points ir_n .

For related results in the unit disc, see Problems 1 and 2 in Chapter 5.]

6.* The results of Exercise 18 extend to the case when γ is assumed merely to be closed, simple, and continuous. The proof, however, requires further ideas.

7.* Applying ideas of Carathéodory, Koebe gave a proof of the Riemann mapping theorem by constructing (more explicitly) a sequence of functions that converges to the desired conformal map.

Starting with a Koebe domain, that is, a simply connected domain $\mathcal{K}_0 \subset \mathbb{D}$ that is not all of $\mathbb{D}$, and which contains the origin, the strategy is to find an injective function f_0 such that $f_0(\mathcal{K}_0) = \mathcal{K}_1$ is a Koebe domain “larger” than $\mathcal{K}_0$. Then, one iterates this process, finally obtaining functions $F_n = f_n \circ \dots \circ f_0 : \mathcal{K}_0 \rightarrow \mathbb{D}$ such that $F_n(\mathcal{K}_0) = \mathcal{K}_{n+1}$ and $\lim F_n = F$ is a conformal map from $\mathcal{K}_0$ to $\mathbb{D}$.

The **inner radius** of a region $\mathcal{K} \subset \mathbb{D}$ that contains the origin is defined by $r_{\mathcal{K}} = \sup\{\rho \geq 0 : D(0, \rho) \subset \mathcal{K}\}$. Also, a holomorphic injection $f : \mathcal{K} \rightarrow \mathbb{D}$ is said to be an **expansion** if $f(0) = 0$ and $|f(z)| > |z|$ for all $z \in \mathcal{K} - \{0\}$.

- (a) Prove that if f is an expansion, then $r_{f(\mathcal{K})} \geq r_{\mathcal{K}}$ and $|f'(0)| > 1$. [Hint: Write $f(z) = zg(z)$ and use the maximum principle to prove that $|f'(0)| = |g(0)| > 1$.]

Suppose we begin with a Koebe domain $\mathcal{K}_0$ and a sequence of expansions $\{f_0, f_1, \dots, f_n, \dots\}$, so that $\mathcal{K}_{n+1} = f_n(\mathcal{K}_n)$ are also Koebe domains. We then define holomorphic maps $F_n : \mathcal{K}_0 \rightarrow \mathbb{D}$ by $F_n = f_n \circ \dots \circ f_0$.

- (b) Prove that for each n , the function F_n is an expansion. Moreover, $F'_n(0) = \prod_{k=0}^n f'_k(0)$, and conclude that $\lim_{n \rightarrow \infty} |f'_n(0)| = 1$. [Hint: Prove that the sequence $\{|F'_n(0)|\}$ has a limit by showing that it is bounded above and monotone increasing. Use the Schwarz lemma.]
- (c) Show that if the sequence is osculating, that is, $r_{\mathcal{K}_n} \rightarrow 1$ as $n \rightarrow \infty$, then $\{F_n\}$ converges uniformly on compact subsets of $\mathcal{K}_0$ to a conformal map $F : \mathcal{K}_0 \rightarrow \mathbb{D}$. [Hint: If $r_{F(\mathcal{K}_0)} \geq 1$ then F is surjective.]

To construct the desired osculating sequence we shall use the automorphisms $\psi_{\alpha} = (\alpha - z)/(1 - \bar{\alpha}z)$.

- (d) Given a Koebe domain $\mathcal{K}$, choose a point $\alpha \in \mathbb{D}$ on the boundary of $\mathcal{K}$ such that $|\alpha| = r_{\mathcal{K}}$, and also choose $\beta \in \mathbb{D}$ such that $\beta^2 = \alpha$. Let S denote the square root of ψ_{α} on $\mathcal{K}$ such that $S(0) = 0$. Why is such a function well defined? Prove that the function $f : \mathcal{K} \rightarrow \mathbb{D}$ defined by $f(z) = \psi_{\beta} \circ S \circ \psi_{\alpha}$

is an expansion. Moreover, show that $|f'(0)| = (1 + r\kappa)/2\sqrt{r\kappa}$. [Hint: To prove that $|f(z)| > |z|$ on $\mathcal{K} - \{0\}$ apply the Schwarz lemma to the inverse function, namely $\psi_\alpha \circ g \circ \psi_\beta$ where $g(z) = z^2$.]

- (e) Use part (d) to construct the desired sequence.

8.* Let f be an injective holomorphic function in the unit disc, with $f(0) = 0$ and $f'(0) = 1$. If we write $f(z) = z + a_2z^2 + a_3z^3 \cdots$, then Problem 1 in Chapter 3 shows that $|a_2| \leq 2$. Bieberbach conjectured that in fact $|a_n| \leq n$ for all $n \geq 2$; this was proved by deBranges. This problem outlines an argument to prove the conjecture under the additional assumption that the coefficients a_n are real.

- (a) Let $z = re^{i\theta}$ with $0 < r < 1$, and show that if $v(r, \theta)$ denotes the imaginary part of $f(re^{i\theta})$, then

$$a_n r^n = \frac{2}{\pi} \int_0^\pi v(r, \theta) \sin n\theta \, d\theta.$$

- (b) Show that for $0 \leq \theta \leq \pi$ and $n = 1, 2, \dots$ we have $|\sin n\theta| \leq n \sin \theta$.
- (c) Use the fact that $a_n \in \mathbb{R}$ to show that $f(\mathbb{D})$ is symmetric with respect to the real axis, and use this fact to show that f maps the upper half-disc into either the upper or lower part of $f(\mathbb{D})$.
- (d) Show that for r small,

$$v(r, \theta) = r \sin \theta [1 + O(r)],$$

and use the previous part to conclude that $v(r, \theta) \sin \theta \geq 0$ for all $0 < r < 1$ and $0 \leq \theta \leq \pi$.

- (e) Prove that $|a_n r^n| \leq nr$, and let $r \rightarrow 1$ to conclude that $|a_n| \leq n$.
- (f) Check that the function $f(z) = z/(1 - z)^2$ satisfies all the hypotheses and that $|a_n| = n$ for all n .

9.* Gauss found a connection between elliptic integrals and the familiar operations of forming arithmetic and geometric means.

We start with any pair (a, b) of numbers that satisfy $a \geq b > 0$, and form the arithmetic and geometric means of a and b , that is,

$$a_1 = \frac{a+b}{2} \quad \text{and} \quad b_1 = (ab)^{1/2}.$$

We then repeat these operations with a and b replaced by a_1 and b_1 . Iterating this process provides two sequences $\{a_n\}$ and $\{b_n\}$ where a_{n+1} and b_{n+1} are the arithmetic and geometric means of a_n and b_n , respectively.

- (a) Prove that the two sequences $\{a_n\}$ and $\{b_n\}$ have a common limit. This limit, which we denote by $M(a, b)$, is called the **arithmetic-geometric mean** of a and b . [Hint: Show that $a \geq a_1 \geq a_2 \geq \cdots \geq a_n \geq b_n \geq \cdots \geq b_1 \geq b$ and $a_n - b_n \leq (a - b)/2^n$.]
- (b) Gauss's identity states that

$$\frac{1}{M(a, b)} = \frac{2}{\pi} \int_0^{\pi/2} \frac{d\theta}{(a^2 \cos^2 \theta + b^2 \sin^2 \theta)^{1/2}}.$$

To prove this relation, show that if $I(a, b)$ denotes the integral on the right-hand side, then it suffices to establish the invariance of I , namely

$$(6) \quad I(a, b) = I\left(\frac{a+b}{2}, (ab)^{1/2}\right).$$

Then, observe that the connection with elliptic integrals takes the form

$$I(a, b) = \frac{1}{a} K(k) = \frac{1}{a} \int_0^1 \frac{dx}{\sqrt{(1-x^2)(1-k^2x^2)}} \quad \text{where } k^2 = 1 - b^2/a^2,$$

and that the relation (6) is a consequence of the identity in (b) of Exercise 24.

9 An Introduction to Elliptic Functions

The form that Jacobi had given to the theory of elliptic functions was far from perfection; its flaws are obvious. At the base we find three fundamental functions sn , cn and dn . These functions do not have the same periods...

In Weierstrass' system, instead of three fundamental functions, there is only one, $\wp(u)$, and it is the simplest of all having the same periods. It has only one double infinity; and finally its definition is so that it does not change when one replaces one system of periods by another equivalent system.

H. Poincaré, 1899

The theory of elliptic functions, which is of interest in several parts of mathematics, initially grew out of the study of elliptic integrals. These can be described generally as integrals of the form $\int R(x, \sqrt{P(x)}) dx$, where R is a rational function and P a polynomial of degree three or four.¹ These integrals arose in computing the arc-length of an ellipse, or of a lemniscate, and in a variety of other problems. Their early study was centered on their special transformation properties and on the discovery of an inherent double-periodicity. We have seen an example of this latter phenomenon in the mapping function of the half-plane to a rectangle taken up in Section 4.5 of the previous chapter.

It was Jacobi who transformed the subject by initiating the systematic study of doubly-periodic functions (called elliptic functions). In this theory, the theta functions he introduced played a decisive role. Weierstrass after him developed another approach, which in its initial steps is simpler and more elegant. It is based on his $\wp$ function, and in this chapter we shall sketch the beginnings of that theory. We will go as far as to glimpse a possible connection with number theory, by considering the Eisenstein series and their expression involving divisor functions. A number of more direct links with combinatorics and number theory arise from the theta

¹The case when P is a quadratic polynomial is essentially that of "circular functions", and can be reduced to the trigonometric functions $\sin x$, $\cos x$, etc.

functions, which we will take up in the next chapter. The remarkable facts we shall see there attest to the great interest of these functions in mathematics. As such they ought to soften the harsh opinion expressed above about the imperfection of Jacobi's theory.

1 Elliptic functions

We are interested in meromorphic functions f on $\mathbb{C}$ that have two periods; that is, there are two non-zero complex numbers ω_1 and ω_2 such that

$$f(z + \omega_1) = f(z) \quad \text{and} \quad f(z + \omega_2) = f(z),$$

for all $z \in \mathbb{C}$. A function with two periods is said to be **doubly periodic**.

The case when ω_1 and ω_2 are linearly dependent over $\mathbb{R}$, that is $\omega_2/\omega_1 \in \mathbb{R}$, is uninteresting. Indeed, Exercise 1 shows that in this case f is either periodic with a simple period (if the quotient ω_2/ω_1 is rational) or f is constant (if ω_2/ω_1 is irrational). Therefore, we make the following assumption: the periods ω_1 and ω_2 are linearly independent over $\mathbb{R}$.

We now describe a normalization that we shall use extensively in this chapter. Let $\tau = \omega_2/\omega_1$. Since τ and $1/\tau$ have imaginary parts of opposite signs, and since τ is not real, we may assume (after possibly interchanging the roles of ω_1 and ω_2) that $\text{Im}(\tau) > 0$. Observe now that the function f has periods ω_1 and ω_2 if and only if the function $F(z) = f(\omega_1 z)$ has periods 1 and τ , and moreover, the function f is meromorphic if and only if F is meromorphic. Also the properties of f are immediately deducible from those of F . We may therefore assume, without loss of generality, that f is a meromorphic function on $\mathbb{C}$ with periods 1 and τ where $\text{Im}(\tau) > 0$.

Successive applications of the periodicity conditions yield

$$(1) \quad f(z + n + m\tau) = f(z) \quad \text{for all integers } n, m \text{ and all } z \in \mathbb{C},$$

and it is therefore natural to consider the lattice in $\mathbb{C}$ defined by

$$\Lambda = \{n + m\tau : n, m \in \mathbb{Z}\}.$$

We say that 1 and τ **generate** Λ (see Figure 1).

Equation (1) says that f is constant under translations by elements of Λ . Associated to the lattice Λ is the **fundamental parallelogram** defined by

$$P_0 = \{z \in \mathbb{C} : z = a + b\tau \text{ where } 0 \leq a < 1 \text{ and } 0 \leq b < 1\}.$$

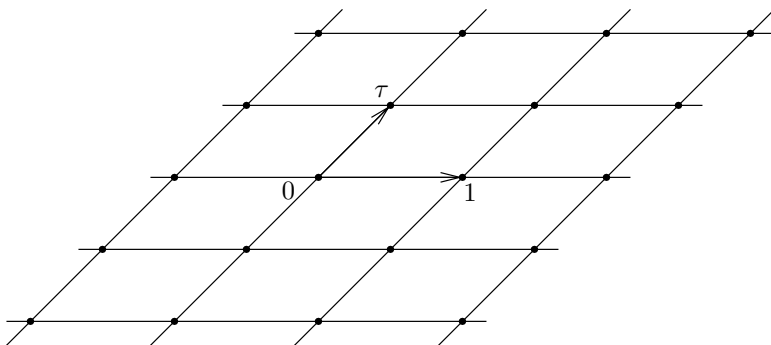


Figure 1. The lattice Λ generated by 1 and τ

The importance of the fundamental parallelogram comes from the fact that f is completely determined by its behavior on P_0 . To see this, we need a definition: two complex numbers z and w are **congruent modulo Λ** if

$$z = w + n + m\tau \quad \text{for some } n, m \in \mathbb{Z},$$

and we write $z \sim w$. In other words, z and w differ by a point in the lattice, $z - w \in \Lambda$. By (1) we conclude that $f(z) = f(w)$ whenever $z \sim w$. If we can show that any point in $z \in \mathbb{C}$ is congruent to a unique point in P_0 then we will have proved that f is completely determined by its values in the fundamental parallelogram. Suppose $z = x + iy$ is given, and write $z = a + b\tau$ where $a, b \in \mathbb{R}$. This is possible since 1 and τ form a basis over the reals of the two-dimensional vector space $\mathbb{C}$. Then choose n and m to be the greatest integers $\leq a$ and $\leq b$, respectively. If we let $w = z - n - m\tau$, then by definition $z \sim w$, and moreover $w = (a - n) + (b - m)\tau$. By construction, it is clear that $w \in P_0$. To prove uniqueness, suppose that w and w' are two points in P_0 that are congruent. If we write $w = a + b\tau$ and $w' = a' + b'\tau$, then $w - w' = (a - a') + (b - b')\tau \in \Lambda$, and therefore both $a - a'$ and $b - b'$ are integers. But since $0 \leq a, a' < 1$, we have $-1 < a - a' < 1$, which then implies $a - a' = 0$. Similarly $b - b' = 0$, and we conclude that $w = w'$.

More generally, a **period parallelogram** P is any translate of the fundamental parallelogram, $P = P_0 + h$ with $h \in \mathbb{C}$ (see Figure 2).

Since we can apply the lemma to $z - h$, we conclude that every point in $\mathbb{C}$ is congruent to a unique point in a given period parallelogram. Therefore, f is uniquely determined by its behavior on any period parallelogram.

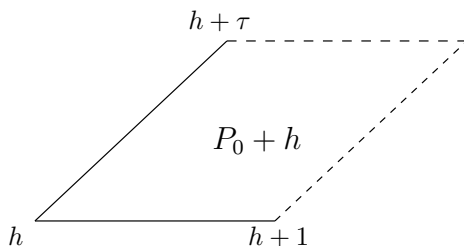


Figure 2. A period parallelogram

Finally, note that Λ and P_0 give rise to a covering (or tiling) of the complex plane

$$(2) \quad \mathbb{C} = \bigcup_{n,m \in \mathbb{Z}} (n + m\tau + P_0),$$

and moreover, this union is disjoint. This is immediate from the facts we just collected and the definition of P_0 . We summarize what we have seen so far.

Proposition 1.1 *Suppose f is a meromorphic function with two periods 1 and τ which generate the lattice Λ . Then:*

- (i) *Every point in $\mathbb{C}$ is congruent to a unique point in the fundamental parallelogram.*
- (ii) *Every point in $\mathbb{C}$ is congruent to a unique point in any given period parallelogram.*
- (iii) *The lattice Λ provides a disjoint covering of the complex plane, in the sense of (2).*
- (iv) *The function f is completely determined by its values in any period parallelogram.*

1.1 Liouville's theorems

We can now see why we assumed from the beginning that f is meromorphic rather than just holomorphic.

Theorem 1.2 *An entire doubly periodic function is constant.*

Proof. The function is completely determined by its values on P_0 and since the closure of P_0 is compact, we conclude that the function is bounded on $\mathbb{C}$, hence constant by Liouville's theorem in Chapter 2.

A non-constant doubly periodic meromorphic function is called an **elliptic function**. Since a meromorphic function can have only finitely many zeros and poles in any large disc, we see that an elliptic function will have only finitely many zeros and poles in any given period parallelogram, and in particular, this is true in the fundamental parallelogram. Of course, nothing excludes f from having a pole or zero on the boundary of P_0 .

As usual, we count poles and zeros with multiplicities. Keeping this in mind we can prove the following theorem.

Theorem 1.3 *The total number of poles of an elliptic function in P_0 is always ≥ 2 .*

In other words, f cannot have only one simple pole. It must have at least two poles, and this does not exclude the case of a single pole of multiplicity ≥ 2 .

Proof. Suppose first that f has no poles on the boundary ∂P_0 of the fundamental parallelogram. By the residue theorem we have

$$\int_{\partial P_0} f(z) dz = 2\pi i \sum \operatorname{res} f,$$

and we contend that the integral is 0. To see this, we simply use the periodicity of f . Note that

$$\int_{\partial P_0} f(z) dz = \int_0^1 f(z) dz + \int_1^{1+\tau} f(z) dz + \int_{1+\tau}^{\tau} f(z) dz + \int_{\tau}^0 f(z) dz,$$

and the integrals over opposite sides cancel out. For instance

$$\begin{aligned} \int_0^1 f(z) dz + \int_{1+\tau}^{\tau} f(z) dz &= \int_0^1 f(z) dz + \int_1^0 f(s + \tau) ds \\ &= \int_0^1 f(z) dz + \int_1^0 f(s) ds \\ &= \int_0^1 f(z) dz - \int_0^1 f(z) dz \\ &= 0, \end{aligned}$$

and similarly for the other pair of sides. Hence $\int_{\partial P_0} f = 0$ and $\sum \operatorname{res} f = 0$. Therefore f must have at least two poles in P_0 .

If f has a pole on ∂P_0 choose a small $h \in \mathbb{C}$ so that if $P = h + P_0$, then f has no poles on ∂P . Arguing as before, we find that f must have at least two poles in P , and therefore the same conclusion holds for P_0 .

The total number of poles (counted according to their multiplicities) of an elliptic function is called its **order**. The next theorem says that elliptic functions have as many zeros as they have poles, if the zeros are counted with their multiplicities.

Theorem 1.4 *Every elliptic function of order m has m zeros in P_0 .*

Proof. Assuming first that f has no zeros or poles on the boundary of P_0 , we know by the argument principle in Chapter 3 that

$$\int_{\partial P_0} \frac{f'(z)}{f(z)} dz = 2\pi i(\mathcal{N}_{\mathfrak{z}} - \mathcal{N}_{\mathfrak{p}})$$

where $\mathcal{N}_{\mathfrak{z}}$ and $\mathcal{N}_{\mathfrak{p}}$ denote the number of zeros and poles of f in P_0 , respectively. By periodicity, we can argue as in the proof of the previous theorem to find that $\int_{\partial P_0} f'/f = 0$, and therefore $\mathcal{N}_{\mathfrak{z}} = \mathcal{N}_{\mathfrak{p}}$.

In the case when a pole or zero of f lies on ∂P_0 it suffices to apply the argument to a translate of P .

As a consequence, if f is elliptic then the equation $f(z) = c$ has as many solutions as the order of f for every $c \in \mathbb{C}$, simply because $f - c$ is elliptic and has as many poles as f .

Despite the rather simple nature of the theorems above, there remains the question of showing that elliptic functions exist. We now turn to a constructive solution of this problem.

1.2 The Weierstrass $\wp$ function

An elliptic function of order two

This section is devoted to the basic example of an elliptic function. As we have seen above, any elliptic function must have at least two poles; we shall in fact construct one whose only singularity will be a double pole at the points of the lattice generated by the periods.

Before looking at the case of doubly-periodic functions, let us first consider briefly functions with only a single period. If one wished to construct a function with period 1 and poles at all the integers, a simple choice would be the sum

$$F(z) = \sum_{n=-\infty}^{\infty} \frac{1}{z+n}.$$

Note that the sum remains unchanged if we replace z by $z + 1$, and the poles are at the integers. However, the series defining F is not absolutely

convergent, and to remedy this problem, we sum symmetrically, that is, we define

$$F(z) = \lim_{N \rightarrow \infty} \sum_{|n| \leq N} \frac{1}{z+n} = \frac{1}{z} + \sum_{n=1}^{\infty} \left[\frac{1}{z+n} + \frac{1}{z-n} \right].$$

On the far right-hand side, we have paired up the terms corresponding to n and $-n$, a trick which makes the quantity in brackets $O(1/n^2)$, and hence the last sum is absolutely convergent. As a consequence, F is meromorphic with poles precisely at the integers. In fact, we proved earlier in Chapter 5 that $F(z) = \pi \cot \pi z$.

There is a second way to deal with the series $\sum_{-\infty}^{\infty} 1/(z+n)$, which is to write it as

$$\frac{1}{z} + \sum_{n \neq 0} \left[\frac{1}{z+n} - \frac{1}{n} \right],$$

where the sum is taken over all non-zero integers. Notice that $1/(z+n) - 1/n = O(1/n^2)$, which makes this series absolutely convergent. Moreover, since

$$\frac{1}{z+n} + \frac{1}{z-n} = \left(\frac{1}{z+n} - \frac{1}{n} \right) + \left(\frac{1}{z-n} - \frac{1}{-n} \right),$$

we get the same sum as before.

In analogy to this, the idea is to mimic the above to produce our first example of an elliptic function. We would like to write it as

$$\sum_{\omega \in \Lambda} \frac{1}{(z+\omega)^2},$$

but again this series does not converge absolutely. There are several approaches to try to make sense of this series (see Problem 1), but the simplest is to follow the second way we dealt with the cotangent series.

To overcome the non-absolute convergence of the series, let Λ^* denote the lattice minus the origin, that is, $\Lambda^* = \Lambda - \{(0,0)\}$, and consider instead the following series:

$$\frac{1}{z^2} + \sum_{\omega \in \Lambda^*} \left[\frac{1}{(z+\omega)^2} - \frac{1}{\omega^2} \right],$$

where we have subtracted the factor $1/\omega^2$ to make the sum converge. The term in brackets is now

$$\frac{1}{(z+\omega)^2} - \frac{1}{\omega^2} = \frac{-z^2 - 2z\omega}{(z+\omega)^2\omega^2} = O\left(\frac{1}{\omega^3}\right) \quad \text{as } |\omega| \rightarrow \infty,$$

and the new series will define a meromorphic function with the desired poles once we have proved the following lemma.

Lemma 1.5 *The two series*

$$\sum_{(n,m) \neq (0,0)} \frac{1}{(|n| + |m|)^r} \quad \text{and} \quad \sum_{n+m\tau \in \Lambda^*} \frac{1}{|n + m\tau|^r}$$

converge if $r > 2$.

Recall that according to the Note at the end of Chapter 7, the question whether a double series converges absolutely is independent of the order of summation. In the present case, we shall first sum in m and then in n .

For the first series, the usual integral comparison can be applied.² For each $n \neq 0$

$$\begin{aligned} \sum_{m \in \mathbb{Z}} \frac{1}{(|n| + |m|)^r} &= \frac{1}{|n|^r} + 2 \sum_{m \geq 1} \frac{1}{(|n| + |m|)^r} \\ &= \frac{1}{|n|^r} + 2 \sum_{k \geq |n|+1} \frac{1}{k^r} \\ &\leq \frac{1}{|n|^r} + 2 \int_{|n|}^{\infty} \frac{dx}{x^r} \\ &\leq \frac{1}{|n|^r} + C \frac{1}{|n|^{r-1}}. \end{aligned}$$

Therefore, $r > 2$ implies

$$\begin{aligned} \sum_{(n,m) \neq (0,0)} \frac{1}{(|n| + |m|)^r} &= \sum_{|m| \neq 0} \frac{1}{|m|^r} + \sum_{|n| \neq 0} \sum_{m \in \mathbb{Z}} \frac{1}{(|n| + |m|)^r} \\ &\leq \sum_{|m| \neq 0} \frac{1}{|m|^r} + \sum_{|n| \neq 0} \left(\frac{1}{|n|^r} + C \frac{1}{|n|^{r-1}} \right) \\ &< \infty. \end{aligned}$$

To prove that the second series also converges, it suffices to show that there is a constant c such that

$$|n| + |m| \leq c|n + \tau m| \quad \text{for all } n, m \in \mathbb{Z}.$$

²We simply use $1/k^r \leq 1/x^r$ when $k-1 \leq x \leq k$; see also the first figure in Chapter 8, Book I.

We use the notation $x \lesssim y$ if there exists a positive constant a such that $x \leq ay$. We also write $x \approx y$ if both $x \lesssim y$ and $y \lesssim x$ hold. Note that for any two positive numbers A and B , one has

$$(A^2 + B^2)^{1/2} \approx A + B.$$

On the one hand $A \leq (A^2 + B^2)^{1/2}$ and $B \leq (A^2 + B^2)^{1/2}$, so that $A + B \leq 2(A^2 + B^2)^{1/2}$. On the other hand, it suffices to square both sides to see that $(A^2 + B^2)^{1/2} \leq A + B$.

The proof that the second series in Lemma 1.5 converges is now a consequence of the following observation:

$$|n| + |m| \approx |n + m\tau| \quad \text{whenever } \tau \in \mathbb{H}.$$

Indeed, if $\tau = s + it$ with $s, t \in \mathbb{R}$ and $t > 0$, then

$$|n + m\tau| = [(n + ms)^2 + (mt)^2]^{1/2} \approx |n + ms| + |mt| \approx |n + ms| + |m|,$$

by the previous observation. Then, $|n + ms| + |m| \approx |n| + |m|$, by considering separately the cases when $|n| \leq 2|m||s|$ and $|n| \geq 2|m||s|$.

Remark. The proof above shows that when $r > 2$ the series $\sum |n + m\tau|^{-r}$ converges uniformly in every half-plane $\text{Im}(\tau) \geq \delta > 0$.

In contrast, when $r = 2$ this series fails to converge (Exercise 3).

With this technical point behind us, we may now return to the definition of the **Weierstrass $\wp$ function**, which is given by the series

$$\begin{aligned} \wp(z) &= \frac{1}{z^2} + \sum_{\omega \in \Lambda^*} \left[\frac{1}{(z + \omega)^2} - \frac{1}{\omega^2} \right] \\ &= \frac{1}{z^2} + \sum_{(n,m) \neq (0,0)} \left[\frac{1}{(z + n + m\tau)^2} - \frac{1}{(n + m\tau)^2} \right]. \end{aligned}$$

We claim that $\wp$ is a meromorphic function with double poles at the lattice points. To see this, suppose that $|z| < R$, and write

$$\wp(z) = \frac{1}{z^2} + \sum_{|\omega| \leq 2R} \left[\frac{1}{(z + \omega)^2} - \frac{1}{\omega^2} \right] + \sum_{|\omega| > 2R} \left[\frac{1}{(z + \omega)^2} - \frac{1}{\omega^2} \right].$$

The term in the second sum is $O(1/|\omega|^3)$ uniformly for $|z| < R$, so by Lemma 1.5 this second sum defines a holomorphic function in $|z| < R$. Finally, note that the first sum exhibits double poles at the lattice points in the disc $|z| < R$.

Observe that because of the insertion of the terms $-1/\omega^2$, it is no longer obvious whether $\wp$ is doubly periodic. Nevertheless this is true, and $\wp$ has all the properties of an elliptic function of order 2. We gather this result in a theorem.

Theorem 1.6 *The function $\wp$ is an elliptic function that has periods 1 and τ , and double poles at the lattice points.*

Proof. It remains only to prove that $\wp$ is periodic with the correct periods. To do so, note that the derivative is given by differentiating the series for $\wp$ termwise so

$$\wp'(z) = -2 \sum_{n,m \in \mathbb{Z}} \frac{1}{(z + n + m\tau)^3}.$$

This accomplishes two things for us. First, the differentiated series converges absolutely whenever z is not a lattice point, by the case $r = 3$ of Lemma 1.5. Second, the differentiation also eliminates the subtraction term $1/\omega^2$; therefore the series for $\wp'$ is clearly periodic with periods 1 and τ , since it remains unchanged after replacing z by $z + 1$ or $z + \tau$.

Hence, there are two constants a and b such that

$$\wp(z + 1) = \wp(z) + a \quad \text{and} \quad \wp(z + \tau) = \wp(z) + b.$$

It is clear from the definition, however, that $\wp$ is even, that is, $\wp(z) = \wp(-z)$, since the sum over $\omega \in \Lambda$ can be replaced by the sum over $-\omega \in \Lambda$. Therefore $\wp(-1/2) = \wp(1/2)$ and $\wp(-\tau/2) = \wp(\tau/2)$, and setting $z = -1/2$ and $z = -\tau/2$, respectively, in the two expressions above proves that $a = b = 0$.

A direct proof of the periodicity of $\wp$ can be given without differentiation; see Exercise 4.

Properties of $\wp$

Several remarks are in order. First, we have already observed that $\wp$ is even, and therefore $\wp'$ is odd. Since $\wp'$ is also periodic with periods 1 and τ , we find that

$$\wp'(1/2) = \wp'(\tau/2) = \wp'\left(\frac{1+\tau}{2}\right) = 0.$$

Indeed, one has, for example,

$$\wp'(1/2) = -\wp'(-1/2) = -\wp'(-1/2 + 1) = -\wp'(1/2).$$

Since $\wp'$ is elliptic and has order 3, the three points $1/2$, $\tau/2$, and $(1 + \tau)/2$ (which are called the **half-periods**) are the only roots of $\wp'$ in the fundamental parallelogram, and they have multiplicity 1. Therefore, if we define

$$\wp(1/2) = e_1, \quad \wp(\tau/2) = e_2, \quad \text{and} \quad \wp\left(\frac{1 + \tau}{2}\right) = e_3,$$

we conclude that the equation $\wp(z) = e_1$ has a double root at $1/2$. Since $\wp$ has order 2, there are no other solutions to the equation $\wp(z) = e_1$ in the fundamental parallelogram. Similarly the equations $\wp(z) = e_2$ and $\wp(z) = e_3$ have only double roots at $\tau/2$ and $(1 + \tau)/2$, respectively. In particular, the three numbers e_1, e_2 , and e_3 are distinct, for otherwise $\wp$ would have at least four roots in the fundamental parallelogram, contradicting the fact that $\wp$ has order 2. From these observations we can prove the following theorem.

Theorem 1.7 *The function $(\wp')^2$ is the cubic polynomial in $\wp$*

$$(\wp')^2 = 4(\wp - e_1)(\wp - e_2)(\wp - e_3).$$

Proof. The only roots of $F(z) = (\wp(z) - e_1)(\wp(z) - e_2)(\wp(z) - e_3)$ in the fundamental parallelogram have multiplicity 2 and are at the points $1/2, \tau/2$, and $(1 + \tau)/2$. Also, $(\wp')^2$ has double roots at these points. Moreover, F has poles of order 6 at the lattice points, and so does $(\wp')^2$ (because $\wp'$ has poles of order 3 there). Consequently $(\wp')^2/F$ is holomorphic and still doubly-periodic, hence this quotient is constant. To find the value of this constant we note that for z near 0, one has

$$\wp(z) = \frac{1}{z^2} + \cdots \quad \text{and} \quad \wp'(z) = \frac{-2}{z^3} + \cdots,$$

where the dots indicate terms of higher order. Therefore the constant is 4, and the theorem is proved.

We next demonstrate the universality of $\wp$ by showing that every elliptic function is a simple combination of $\wp$ and $\wp'$.

Theorem 1.8 *Every elliptic function f with periods 1 and τ is a rational function of $\wp$ and $\wp'$.*

The theorem will be an easy consequence of the following version of it.

Lemma 1.9 *Every even elliptic function F with periods 1 and τ is a rational function of $\wp$.*

Proof. If F has a zero or pole at the origin it must be of even order, since F is an even function. As a consequence, there exists an integer m so that $F\wp^m$ has no zero or pole at the lattice points. We may therefore assume that F itself has no zero or pole on Λ .

Our immediate goal is to use $\wp$ to construct a doubly-periodic function G with precisely the same zeros and poles as F . To achieve this, we recall that $\wp(z) - \wp(a)$ has a single zero of order 2 if a is a half-period, and two distinct zeros at a and $-a$ otherwise. We must therefore carefully count the zeros and poles of F .

If a is a zero of F , then so is $-a$, since F is even. Moreover, a is congruent to $-a$ if and only if it is a half-period, in which case the zero is of even order. Therefore, if the points $a_1, -a_1, \dots, a_m, -a_m$ counted with multiplicities³ describe all the zeros of F , then

$$[\wp(z) - \wp(a_1)] \cdots [\wp(z) - \wp(a_m)]$$

has precisely the same roots as F . A similar argument, where $b_1, -b_1, \dots, b_m, -b_m$ (with multiplicities) describe all the poles of F , then shows that

$$G(z) = \frac{[\wp(z) - \wp(a_1)] \cdots [\wp(z) - \wp(a_m)]}{[\wp(z) - \wp(b_1)] \cdots [\wp(z) - \wp(b_m)]}$$

is periodic and has the same zeros and poles as F . Therefore, F/G is holomorphic and doubly-periodic, hence constant. This concludes the proof of the lemma.

To prove the theorem, we first recall that $\wp$ is even while $\wp'$ is odd. We then write f as a sum of an even and an odd function,

$$f(z) = f_{\text{even}}(z) + f_{\text{odd}}(z),$$

where in fact

$$f_{\text{even}}(z) = \frac{f(z) + f(-z)}{2} \quad \text{and} \quad f_{\text{odd}}(z) = \frac{f(z) - f(-z)}{2}.$$

Then, since $f_{\text{odd}}/\wp'$ is even, it is clear from the lemma applied to f_{even} and $f_{\text{odd}}/\wp'$ that f is a rational function of $\wp$ and $\wp'$.

³If a_j is not a half-period, then a_j and $-a_j$ have the multiplicity of F at these points. If a_j is a half-period, then a_j and $-a_j$ are congruent and each has multiplicity half of the multiplicity of F at this point.

2 The modular character of elliptic functions and Eisenstein series

We shall now study the modular character of elliptic functions, that is, their dependence on τ .

Recall the normalization we made at the beginning of the chapter. We started with two periods ω_1 and ω_2 linearly that are independent over $\mathbb{R}$, and we defined $\tau = \omega_2/\omega_1$. We could then assume that $\text{Im}(\tau) > 0$, and also that the two periods are 1 and τ . Next, we considered the lattice generated by 1 and τ and constructed the function $\wp$, which is elliptic of order 2 with periods 1 and τ . Since the construction of $\wp$ depends on τ , we could write $\wp_\tau$ instead. This leads us to change our point of view and think of $\wp_\tau(z)$ primarily as a function of τ . This approach yields many interesting new insights.

Our considerations are guided by the following observations. First, since 1 and τ generate the periods of $\wp_\tau(z)$, and 1 and $\tau + 1$ generate the same periods, we can expect a close relationship between $\wp_\tau(z)$ and $\wp_{\tau+1}(z)$. In fact, it is easy to see that they are identical. Second, since $\tau = \omega_2/\omega_1$, by the normalization imposed at the beginning of Section 1, we see that $-1/\tau = -\omega_1/\omega_2$ (with $\text{Im}(-1/\tau) > 0$). This corresponds essentially to an interchange of the two periods ω_1 and ω_2 , and thus we can also expect an intimate connection between $\wp_\tau$ and $\wp_{-1/\tau}$. In fact, it is easy to verify that $\wp_{-1/\tau}(z) = \tau^2 \wp_\tau(\tau z)$.

So we are led to consider the group of transformations of the upper half-plane $\text{Im}(\tau) > 0$, generated by the two transformations $\tau \mapsto \tau + 1$ and $\tau \mapsto -1/\tau$. This group is called the **modular group**. On the basis of what we said, it can be expected that all quantities intrinsically attached to $\wp_\tau(z)$ reflect the above transformations. We see this clearly when we consider the Eisenstein series.

2.1 Eisenstein series

The **Eisenstein series** of order k is defined by

$$E_k(\tau) = \sum_{(n,m) \neq (0,0)} \frac{1}{(n + m\tau)^k},$$

whenever k is an integer ≥ 3 and τ is a complex number with $\text{Im}(\tau) > 0$. If Λ is the lattice generated by 1 and τ , and if we write $\omega = n + m\tau$, then another expression for the Eisenstein series is $\sum_{\omega \in \Lambda^*} 1/\omega^k$.

Theorem 2.1 *Eisenstein series have the following properties:*

- (i) The series $E_k(\tau)$ converges if $k \geq 3$, and is holomorphic in the upper half-plane.
- (ii) $E_k(\tau) = 0$ if k is odd.
- (iii) $E_k(\tau)$ satisfies the following transformation relations:

$$E_k(\tau + 1) = E_k(\tau) \quad \text{and} \quad E_k(\tau) = \tau^{-k} E_k(-1/\tau).$$

The last property is sometimes referred to as the **modular** character of the Eisenstein series. We shall return to these and other modular identities in the next chapter.

Proof. By Lemma 1.5 and the remark after it, the series $E_k(\tau)$ converges absolutely and uniformly in every half-plane $\text{Im}(\tau) \geq \delta > 0$, whenever $k \geq 3$; hence $E_k(\tau)$ is holomorphic in the upper half-plane $\text{Im}(\tau) > 0$.

By symmetry, replacing n and m by $-n$ and $-m$, we see that whenever k is odd the Eisenstein series is identically zero.

Finally, the fact that $E_k(\tau)$ is periodic of period 1 is clear from the fact that $n + m(\tau + 1) = n + m + m\tau$, and that we can rearrange the sum by replacing $n + m$ by n . Also, we have

$$(n + m(-1/\tau))^k = \tau^{-k} (n\tau - m)^k,$$

and again we can rearrange the sum, this time replacing $(-m, n)$ by (n, m) . Conclusion (iii) then follows.

Remark. Because of the second property, some authors define the Eisenstein series of order k to be $\sum_{(n,m) \neq (0,0)} 1/(n + m\tau)^{2k}$, possibly also with a constant factor in front.

The connection of the E_k with the Weierstrass $\wp$ function arises when we investigate the series expansion of $\wp$ near 0.

Theorem 2.2 For z near 0, we have

$$\begin{aligned} \wp(z) &= \frac{1}{z^2} + 3E_4 z^2 + 5E_6 z^4 + \cdots \\ &= \frac{1}{z^2} + \sum_{k=1}^{\infty} (2k+1) E_{2k+2} z^{2k}. \end{aligned}$$

Proof. From the definition of $\wp$, if we note that we may replace ω by $-\omega$ without changing the sum, we have

$$\wp(z) = \frac{1}{z^2} + \sum_{\omega \in \Lambda^*} \left[\frac{1}{(z + \omega)^2} - \frac{1}{\omega^2} \right] = \frac{1}{z^2} + \sum_{\omega \in \Lambda^*} \left[\frac{1}{(z - \omega)^2} - \frac{1}{\omega^2} \right],$$

where $\omega = n + m\tau$. The identity

$$\frac{1}{(1-w)^2} = \sum_{\ell=0}^{\infty} (\ell+1)w^{\ell}, \quad \text{for } |w| < 1,$$

which follows from differentiating the geometric series, implies that for all small z

$$\frac{1}{(z-\omega)^2} = \frac{1}{\omega^2} \sum_{\ell=0}^{\infty} (\ell+1) \left(\frac{z}{\omega}\right)^{\ell} = \frac{1}{\omega^2} + \frac{1}{\omega^2} \sum_{\ell=1}^{\infty} (\ell+1) \left(\frac{z}{\omega}\right)^{\ell}.$$

Therefore

$$\begin{aligned} \wp(z) &= \frac{1}{z^2} + \sum_{\omega \in \Lambda^*} \sum_{\ell=1}^{\infty} (\ell+1) \frac{z^{\ell}}{\omega^{\ell+2}} \\ &= \frac{1}{z^2} + \sum_{\ell=1}^{\infty} (\ell+1) \left(\sum_{\omega \in \Lambda^*} \frac{1}{\omega^{\ell+2}} \right) z^{\ell} \\ &= \frac{1}{z^2} + \sum_{\ell=1}^{\infty} (\ell+1) E_{\ell+2} z^{\ell} \\ &= \frac{1}{z^2} + \sum_{k=1}^{\infty} (2k+1) E_{2k+2} z^{2k}, \end{aligned}$$

where we have used the fact that $E_{\ell+2} = 0$ whenever ℓ is odd.

From this theorem, we obtain the following three expansions for z near 0:

$$\begin{aligned} \wp'(z) &= \frac{-2}{z^3} + 6E_4 z + 20E_6 z^3 + \cdots, \\ (\wp'(z))^2 &= \frac{4}{z^6} - \frac{24E_4}{z^2} - 80E_6 + \cdots, \\ (\wp(z))^3 &= \frac{1}{z^6} + \frac{9E_4}{z^2} + 15E_6 + \cdots. \end{aligned}$$

From these, one sees that the difference $(\wp'(z))^2 - 4(\wp(z))^3 + 60E_4\wp(z) + 140E_6$ is holomorphic near 0, and in fact equal to 0 at the origin. Since this difference is also doubly periodic, we conclude by Theorem 1.2 that it is constant, and hence identically 0. This proves the following corollary.

Corollary 2.3 *If $g_2 = 60E_4$ and $g_3 = 140E_6$, then*

$$(\wp')^2 = 4\wp^3 - g_2\wp - g_3.$$

Note that this identity is another version of Theorem 1.7, and it allows one to express the symmetric functions of the e_j 's in terms of the Eisenstein series.

2.2 Eisenstein series and divisor functions

We will describe now the link between Eisenstein series and some number-theoretic quantities. This relation comes about if we consider the Fourier coefficients in the Fourier expansion of the periodic function $E_k(\tau)$. Equivalently, we can write $\mathcal{E}(z) = E_k(\tau)$ with $z = e^{2\pi i\tau}$, and investigate the Laurent expansion of $\mathcal{E}$ as a function of z .

We begin with a lemma.

Lemma 2.4 *If $k \geq 2$ and $\text{Im}(\tau) > 0$, then*

$$\sum_{n=-\infty}^{\infty} \frac{1}{(n+\tau)^k} = \frac{(-2\pi i)^k}{(k-1)!} \sum_{\ell=1}^{\infty} \ell^{k-1} e^{2\pi i\tau\ell}.$$

Proof. This identity follows from applying the Poisson summation formula to $f(z) = 1/(z+\tau)^k$; see Exercise 7 in Chapter 4.

An alternate proof consists of noting that it first suffices to establish the formula for $k = 2$, since the other cases are then obtained by differentiating term by term. To prove this special case, we differentiate the formula for the cotangent derived in Chapter 5

$$\sum_{n=-\infty}^{\infty} \frac{1}{n+\tau} = \pi \cot \pi\tau.$$

This yields

$$\sum_{n=-\infty}^{\infty} \frac{1}{(n+\tau)^2} = \frac{\pi^2}{\sin^2(\pi\tau)}.$$

Now use Euler's formula for the sine and the fact that

$$\sum_{r=1}^{\infty} r w^r = \frac{w}{(1-w)^2} \quad \text{with } w = e^{2\pi i\tau}$$

to obtain the desired result.

As a consequence of this lemma, we can draw a connection between the Eisenstein series, the zeta function, and the divisor functions. The

divisor function $\sigma_\ell(r)$ that arises here is defined as the sum of the ℓ^{th} powers of the divisors of r , that is,

$$\sigma_\ell(r) = \sum_{d|r} d^\ell.$$

Theorem 2.5 *If $k \geq 4$ is even, and $\text{Im}(\tau) > 0$, then*

$$E_k(\tau) = 2\zeta(k) + \frac{2(-1)^{k/2}(2\pi)^k}{(k-1)!} \sum_{r=1}^{\infty} \sigma_{k-1}(r) e^{2\pi i \tau r}.$$

Proof. First observe that $\sigma_{k-1}(r) \leq rr^{k-1} = r^k$. If $\text{Im}(\tau) = t$, then whenever $t \geq t_0$ we have $|e^{2\pi i r \tau}| \leq e^{-2\pi r t_0}$, and we see that the series in the theorem is absolutely convergent in any half-plane $t \geq t_0$, by comparison with $\sum_{r=1}^{\infty} r^k e^{-2\pi r t_0}$. To establish the formula, we use the definition of E_k , that of ζ , the fact that k is even, and the previous lemma (with τ replaced by $m\tau$) to get successively

$$\begin{aligned} E_k(\tau) &= \sum_{(n,m) \neq (0,0)} \frac{1}{(n+m\tau)^k} \\ &= \sum_{n \neq 0} \frac{1}{n^k} + \sum_{m \neq 0} \sum_{n=-\infty}^{\infty} \frac{1}{(n+m\tau)^k} \\ &= 2\zeta(k) + \sum_{m \neq 0} \sum_{n=-\infty}^{\infty} \frac{1}{(n+m\tau)^k} \\ &= 2\zeta(k) + 2 \sum_{m>0} \sum_{n=-\infty}^{\infty} \frac{1}{(n+m\tau)^k} \\ &= 2\zeta(k) + 2 \sum_{m>0} \frac{(-2\pi i)^k}{(k-1)!} \sum_{\ell=1}^{\infty} \ell^{k-1} e^{2\pi i m \tau \ell} \\ &= 2\zeta(k) + \frac{2(-1)^{k/2}(2\pi)^k}{(k-1)!} \sum_{m>0} \sum_{\ell=1}^{\infty} \ell^{k-1} e^{2\pi i m \tau \ell} \\ &= 2\zeta(k) + \frac{2(-1)^{k/2}(2\pi)^k}{(k-1)!} \sum_{r=1}^{\infty} \sigma_{k-1}(r) e^{2\pi i \tau r}. \end{aligned}$$

This proves the desired formula.

Finally, we turn to the *forbidden* case $k = 2$. The series we have in mind $\sum_{(n,m) \neq (0,0)} 1/(n+m\tau)^2$ no longer converges absolutely, but we

seek to give it a meaning anyway. We define

$$F(\tau) = \sum_m \left(\sum_n \frac{1}{(n + m\tau)^2} \right)$$

summed in the indicated order with $(n, m) \neq (0, 0)$. The argument given in the above theorem proves that the double sum converges, and in fact has the expected expression.

Corollary 2.6 *The double sum defining F converges in the indicated order. We have*

$$F(\tau) = 2\zeta(2) - 8\pi^2 \sum_{r=1}^{\infty} \sigma(r) e^{2\pi i r \tau},$$

where $\sigma(r) = \sum_{d|r} d$ is the sum of the divisors of r .

It can be seen that $F(-1/\tau)\tau^{-2}$ does *not* equal $F(\tau)$, and this is the same as saying that the double series for F gives a different value ($\tilde{F}$, the reverse of F) when we sum first in m and then in n . It turns out that nevertheless the **forbidden Eisenstein series** $F(\tau)$ can be used in a crucial way in the proof of the celebrated theorem about representing an integer as the sum of four squares. We turn to these matters in the next chapter.

3 Exercises

1. Suppose that a meromorphic function f has two periods ω_1 and ω_2 , with $\omega_2/\omega_1 \in \mathbb{R}$.

- (a) Suppose ω_2/ω_1 is rational, say equal to p/q , where p and q are relatively prime integers. Prove that as a result the periodicity assumption is equivalent to the assumption that f is periodic with the simple period $\omega_0 = \frac{1}{q}\omega_1$. [Hint: Since p and q are relatively prime, there exist integers m and n such that $mq + np = 1$ (Corollary 1.3, Chapter 8, Book I).]
- (b) If ω_2/ω_1 is irrational, then f is constant. To prove this, use the fact that $\{m - n\tau\}$ is dense in $\mathbb{R}$ whenever τ is irrational and m, n range over the integers.

2. Suppose that $a_1, \dots, a_r$ and $b_1, \dots, b_r$ are the zeros and poles, respectively, in the fundamental parallelogram of an elliptic function f . Show that

$$a_1 + \dots + a_r - b_1 - \dots - b_r = n\omega_1 + m\omega_2$$

for some integers n and m .

[Hint: If the boundary of the parallelogram contains no zeros or poles, simply integrate $zf'(z)/f(z)$ over that boundary, and observe that the integral of $f'(z)/f(z)$ over a side is an integer multiple of $2\pi i$. If there are zeros or poles on the side of the parallelogram, translate it by a small amount to reduce the problem to the first case.]

3. In contrast with the result in Lemma 1.5, prove that the series

$$\sum_{n+m\tau \in \Lambda^*} \frac{1}{|n+m\tau|^2} \quad \text{where } \tau \in \mathbb{H}$$

does not converge. In fact, show that

$$\sum_{1 \leq n^2 + m^2 \leq R^2} 1/(n^2 + m^2) = 2\pi \log R + O(1) \quad \text{as } R \rightarrow \infty.$$

4. By rearranging the series

$$\frac{1}{z^2} + \sum_{\omega \in \Lambda^*} \left[\frac{1}{(z+\omega)^2} - \frac{1}{\omega^2} \right],$$

show directly, without differentiation, that $\wp(z+\omega) = \wp(z)$ whenever $\omega \in \Lambda$.

[Hint: For R sufficiently large, note that $\wp(z) = \wp^R(z) + O(1/R)$, where $\wp^R(z) = z^{-2} + \sum_{0 < |\omega| < R} ((z+\omega)^{-2} - \omega^{-2})$. Next, observe that both $\wp^R(z+1) - \wp^R(z)$ and $\wp^R(z+\tau) - \wp^R(z)$ are $O(\sum_{R-c < |\omega| < R+c} |\omega|^{-2}) = O(1/R)$.]

5. Let $\sigma(z)$ be the canonical product

$$\sigma(z) = z \prod_{j=1}^{\infty} E_2(z/\tau_j),$$

where τ_j is an enumeration of the periods $\{n+m\tau\}$ with $(n,m) \neq (0,0)$, and $E_2(z) = (1-z)e^{z+z^2/2}$.

(a) Show that $\sigma(z)$ is an entire function of order 2 that has simple zeros at all the periods $n+m\tau$, and vanishes nowhere else.

(b) Show that

$$\frac{\sigma'(z)}{\sigma(z)} = \frac{1}{z} + \sum_{(n,m) \neq (0,0)} \left[\frac{1}{z-n-m\tau} + \frac{1}{n+m\tau} + \frac{z}{(n+m\tau)^2} \right],$$

and that this series converges whenever z is not a lattice point.

(c) Let $L(z) = -\sigma'(z)/\sigma(z)$. Then

$$L'(z) = \frac{(\sigma'(z))^2 - \sigma(z)\sigma''(z)}{(\sigma(z))^2} = \wp(z).$$

6. Prove that $\wp''$ is a quadratic polynomial in $\wp$.

7. Setting $\tau = 1/2$ in the expression

$$\sum_{m=-\infty}^{\infty} \frac{1}{(m+\tau)^2} = \frac{\pi^2}{\sin^2(\pi\tau)},$$

deduce that

$$\sum_{m \geq 1, m \text{ odd}} \frac{1}{m^2} = \frac{\pi^2}{8} \quad \text{and} \quad \sum_{m \geq 1} \frac{1}{m^2} = \frac{\pi^2}{6} = \zeta(2).$$

Similarly, using $\sum 1/(m+\tau)^4$ deduce that

$$\sum_{m \geq 1, m \text{ odd}} \frac{1}{m^4} = \frac{\pi^4}{96} \quad \text{and} \quad \sum_{m \geq 1} \frac{1}{m^4} = \frac{\pi^4}{90} = \zeta(4).$$

These results were already obtained using Fourier series in the exercises at the end of Chapters 2 and 3 in Book I.

8. Let

$$E_4(\tau) = \sum_{(n,m) \neq (0,0)} \frac{1}{(n+m\tau)^4}$$

be the Eisenstein series of order 4.

(a) Show that $E_4(\tau) \rightarrow \pi^4/45$ as $\text{Im}(\tau) \rightarrow \infty$.

(b) More precisely,

$$\left| E_4(\tau) - \frac{\pi^4}{45} \right| \leq ce^{-2\pi t} \quad \text{if } \tau = x + it \text{ and } t \geq 1.$$

(c) Deduce that

$$\left| E_4(\tau) - \tau^{-4} \frac{\pi^4}{45} \right| \leq ct^{-4} e^{-2\pi/t} \quad \text{if } \tau = it \text{ and } 0 < t \leq 1.$$

4 Problems

1. Besides the approach in Section 1.2, there are several alternate ways of dealing with the sum $\sum 1/(z + \omega)^2$, where $\omega = n + m\tau$. For example, one may sum either (a) circularly, (b) first in n then in m , (c) or first in m then in n .

(a) Prove that if $z \notin \Lambda$, then

$$\lim_{R \rightarrow \infty} \sum_{n^2 + m^2 \leq R^2} \frac{1}{(z + n + m\tau)^2} = S_1(z)$$

exists and $S_1(z) = \wp(z) + c_1$.

(b) Similarly,

$$\sum_m \left(\sum_n \frac{1}{(z + n + m\tau)^2} \right) = S_2(z)$$

exists and $S_2(z) = \wp(z) + c_2$, where $c_2 = F(\tau)$, and F is the forbidden Eisenstein series.

(c) Also

$$\sum_n \left(\sum_m \frac{1}{(z + n + m\tau)^2} \right) = S_3(z)$$

exists with $S_3(z) = \wp(z) + c_3$, and $c_3 = \tilde{F}(\tau)$, the reverse of F .

[Hint: To prove (a), it suffices to show that $\lim_{R \rightarrow \infty} \sum_{1 \leq n^2 + m^2 \leq R^2} 1/(n + m\tau)^2 = c_1$

exists. This is proved by a comparison with $\int_{1 \leq x^2 + y^2 \leq R^2} \frac{dx}{(x + y\tau)^2} = I(R)$. It can be shown that $I(R) = 0$, which follows because $(x + y\tau)^{-2} = -(\partial/\partial x)(x + y\tau)^{-1}$.]

2. Show that

$$\wp(z) = c + \pi^2 \sum_{m=-\infty}^{\infty} \frac{1}{\sin^2((z + m\tau)\pi)}$$

where c is an appropriate constant. In fact, by part (b) of the previous problem $c = -F(\tau)$.

3.* Suppose Ω is a simply connected domain that excludes the three roots of the polynomial $4z^3 - g_2z - g_3$. For $\omega_0 \in \Omega$ and ω_0 fixed, define the function I on Ω by

$$I(\omega) = \int_{\omega_0}^{\omega} \frac{dz}{\sqrt{4z^3 - g_2z - g_3}} \quad \omega \in \Omega.$$

Then the function I has an inverse given by $\wp(z + \alpha)$ for some constant α ; that is,

$$I(\wp(z + \alpha)) = z$$